Asymmetric Information

Martin J. Conyon

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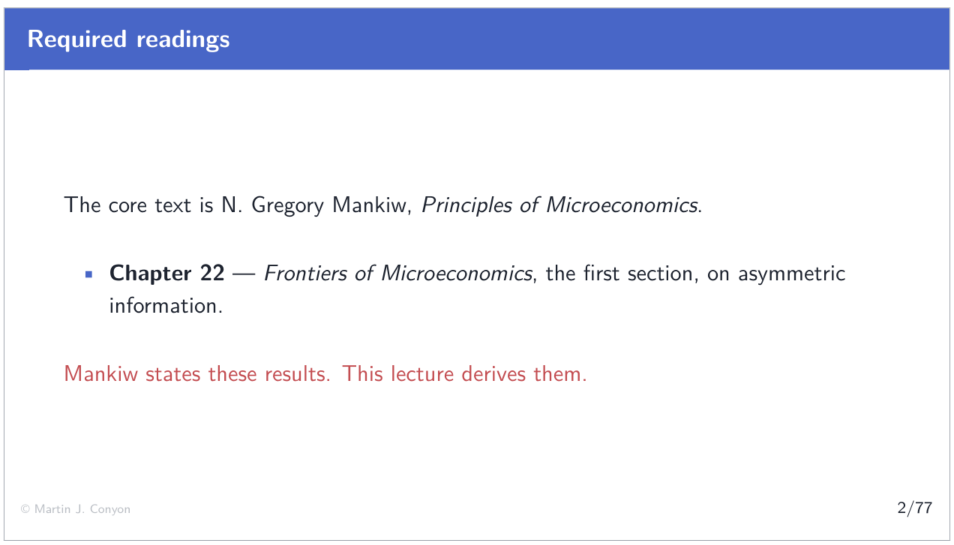
A companion to the lecture slides. Each slide that hides a step is reproduced here and worked through line by line.

Before we begin

Required readings



Slide 2

Working through it

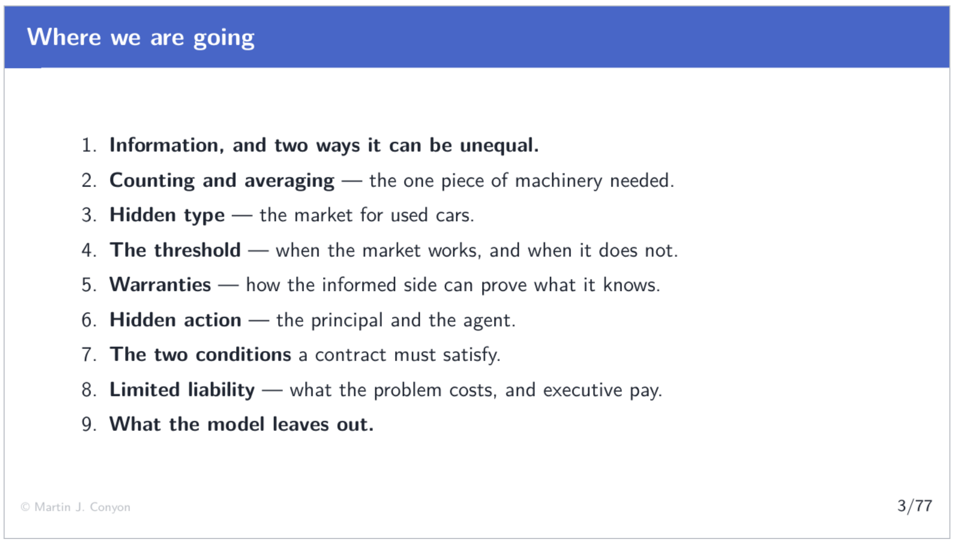
What the chapter contains. Mankiw's chapter 22 is titled *Frontiers of Microeconomics*, and its first section covers asymmetric information: hidden actions and moral hazard, hidden characteristics and adverse selection, then signaling and screening.

What it does not contain. Any derivation. The used-car result is asserted in a paragraph; the incentive problem is described in words. There is no threshold, no condition on a contract, and no arithmetic anywhere in the section.

Why that matters for how you read it. The chapter is a good survey and it will not tell you *why* any of it is true. Read it for the vocabulary and the range of applications — insurance, labor markets, education — and take the derivations from here.

Where chapter 22 sits. It is the last chapter of the book, positioned as a survey of topics students might meet later. Most courses never reach it. The material is not difficult, and none of it requires anything beyond counting and two linear inequalities.

Where we are going



Slide 3

Working through it

The shape of the lecture. Two halves of four parts each, with one part of machinery at the front.

Part 2 is arithmetic and contains no economics. It builds the weighted average from counting. Everything after it uses that and nothing harder.

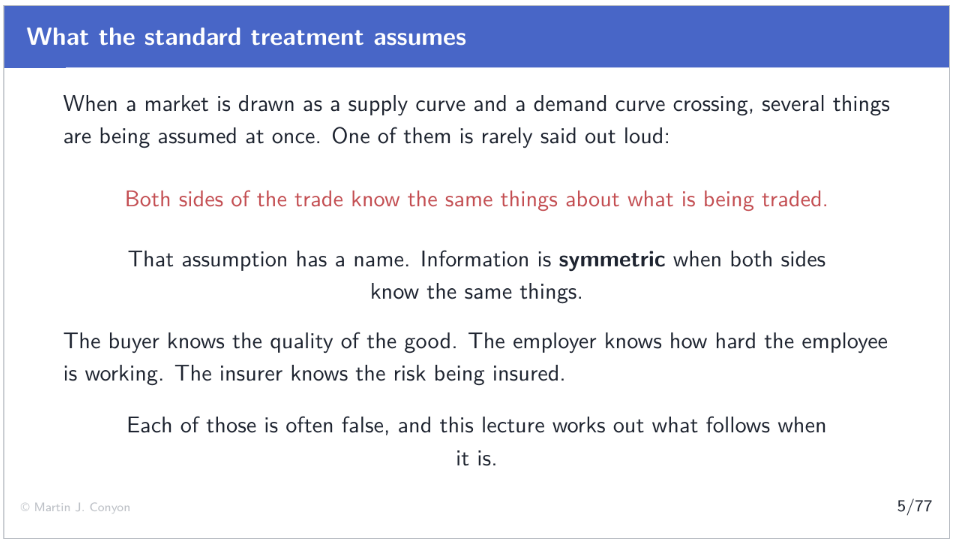
Parts 3 to 5 are hidden type. A market is set up in which every trade is worth making; a condition is derived under which the good ones stop happening; and then the response — a warranty — is derived as well, so the half ends with what markets do about the problem rather than with the problem.

Parts 6 to 8 are hidden action. A contract cannot condition on effort because effort is not observed, so it conditions on output instead. Two inequalities follow, and their interaction with the fact that wages cannot be negative is what the problem costs.

Part 9 states the assumptions. Four of them do real work and each fails somewhere.

1. Part 1: Information, and Two Ways It Can Be Unequal

What the standard treatment assumes



Slide 5

Working through it

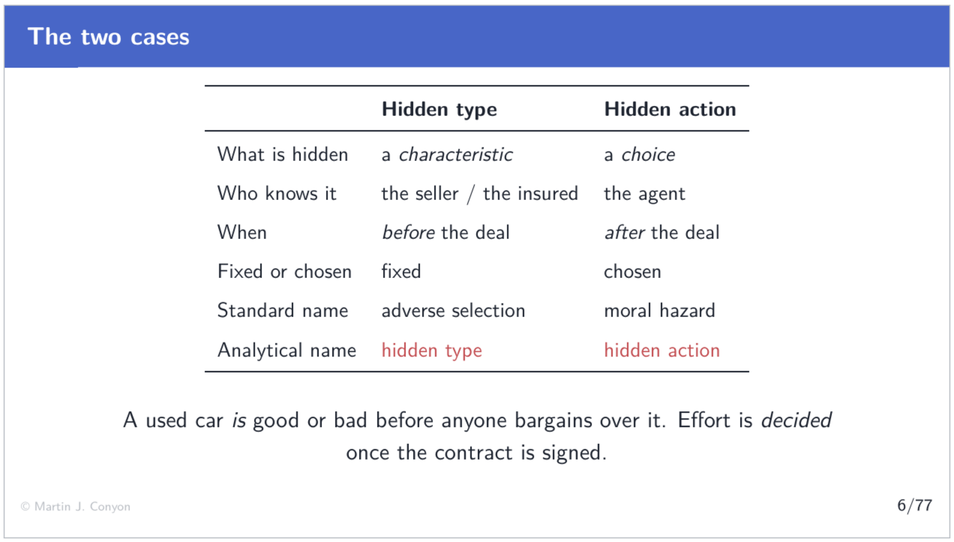
The assumption, stated exactly. Both parties to a trade have the same information about the object of the trade. Economists call this **symmetric information**: not that everything is known, but that whatever is known is known by both.

Where it is buried. It sits inside the demand curve. A demand curve records what buyers are willing to pay for a good, and that willingness depends on what the good is. If buyers cannot tell one unit from another, then there is no single good being demanded, and the curve is not well defined until you say what buyers believe they are getting.

Three markets where it fails, and they are not marginal cases. Second-hand goods, where the seller has driven the car and the buyer has not. Labor, where the employee knows the effort being supplied and the employer sees only the result. Insurance, where the applicant knows the risk far better than the insurer.

What this lecture is not claiming. Not that anyone lies. Nothing in what follows requires dishonesty at any point, and the failure occurs even when every party is entirely truthful. The problem is that some things are not observable, not that people misreport them.

The two cases



Slide 6

Working through it

One distinction, four consequences. Everything in the table follows from whether the hidden thing is a characteristic or a choice.

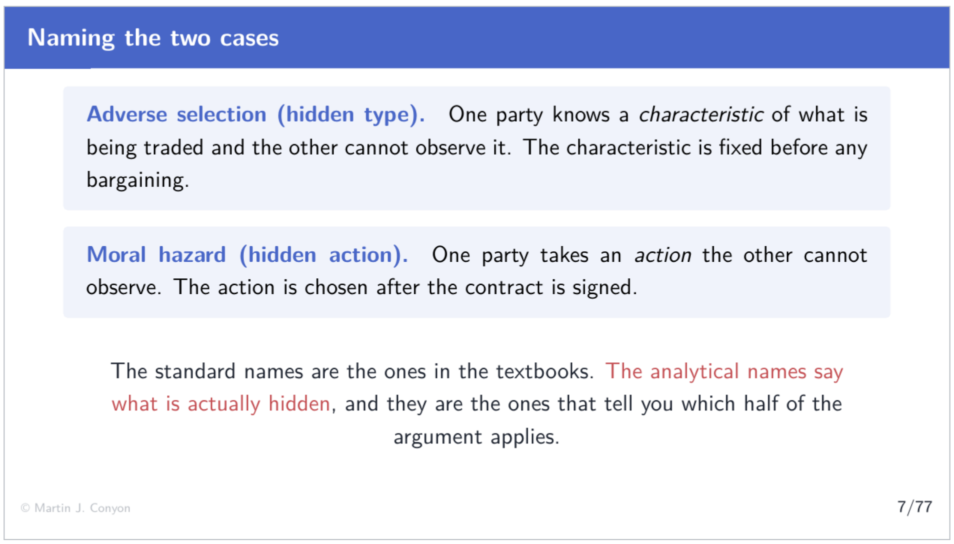
Hidden type. The car's condition exists before the buyer appears. No contract can change it. The problem is that the buyer cannot tell which car is in front of them, so a single price must serve for cars of different worth – and that price is wrong for both.

Hidden action. The manager's effort does not exist until the manager supplies it. A contract signed now can change it, by making the manager's pay depend on something correlated with it. The problem is that the contract cannot name effort directly, because nobody can verify what was supplied.

Why the names are unhelpful and used anyway. "Adverse selection" names a consequence rather than a cause: the selection of who chooses to trade is adverse to the uninformed party. "Moral hazard" comes from nineteenth-century insurance and carries a suggestion of bad character that the economics does not require — responding to incentives is not a moral failing. Both terms are standard, so both are used, but **hidden type** and **hidden action** say what is actually going on.

The two can occur together. A driver knows both their own carefulness (a characteristic) and how carefully they will drive once insured (a choice). Insurance markets face both problems at once. This lecture takes them one at a time.

Naming the two cases



Slide 7

Working through it

Two vocabularies for the same distinction. The table on the previous page sets the two cases side by side. This page fixes the words, because both pairs of names are in use and a student will meet them all.

Adverse selection, or **hidden type**. One party knows a characteristic of the object of trade and the other cannot observe it. The characteristic is *fixed before any bargaining*: the car is already good or bad, the applicant already carries the condition. No contract can alter it.

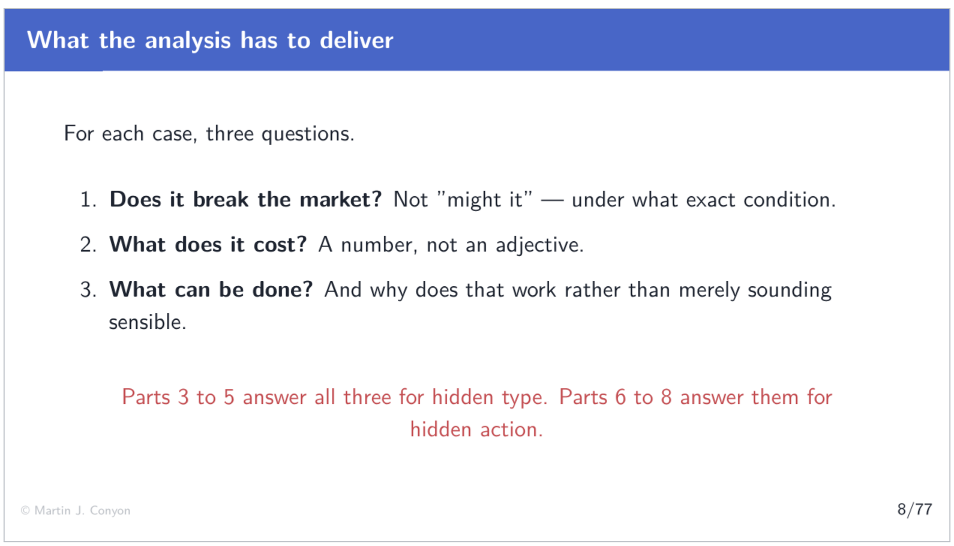
Moral hazard, or **hidden action**. One party takes an action the other cannot observe.

The action is *chosen after the contract is signed*: the effort is supplied, or the care is taken, once the terms are known. A contract written now can change it.

Why the standard names are kept despite being poor. "Adverse selection" names a consequence — the selection of who chooses to trade is adverse to the uninformed party — rather than the cause. "Moral hazard" comes from nineteenth-century insurance and implies bad character that the analysis nowhere requires. Both are what the textbooks and the market use, so both are worth knowing.

Why the analytical names are the working ones. *Hidden type* and *hidden action* say what is hidden, and that is the only thing you need in order to know which apparatus applies. Faced with a new case, ask: is the hidden thing a characteristic or a choice? That question decides everything else.

What the analysis has to deliver



Slide 8

Working through it

Why set the questions out in advance. Because the answers are the structure of the lecture, and it is easier to follow an argument when you know what it is trying to establish.

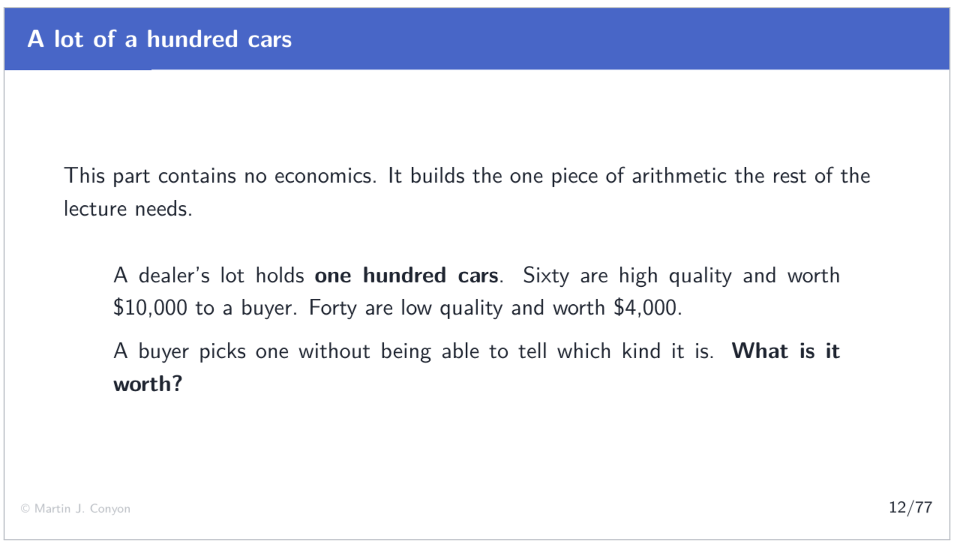
Under what condition. A claim that a market "might fail" is not worth much. What is worth having is a threshold: a number such that above it the market works and below it the good side of it withdraws. Part 4 derives one, in terms of the underlying valuations rather than for one set of figures.

A number, not an adjective. "Significant welfare loss" is a phrase. $120,000 on a lot of a hundred cars is a measurement, and it can be compared against the cost of doing something about it.

Why the remedy works. This is where most treatments stop short. It is easy to say that warranties help. The question is why a warranty is *credible* — why a seller with a bad car cannot simply offer the same warranty and pocket the higher price. Part 5 answers that, and the answer is a condition on numbers rather than an appeal to honesty.

2. Part 2: Counting and Averaging

A lot of a hundred cars



Slide 12

Working through it

Why this part exists. What follows needs a weighted average, and the notation for one — an expectation — is not assumed here. It is built from counting, which needs nothing beyond arithmetic.

Note the framing. Sixty cars out of a hundred, not "a probability of nought point six". The two are the same fact, and the first requires no apparatus. Everywhere in this lecture a chance is a count out of a hundred.

The question. A buyer who cannot tell the two kinds apart is not buying a high-quality car or a low-quality car. They are buying *a car from this lot*, and the question is what that is worth. Four steps.

Step 1 — what is the whole lot worth?

Step 1 — what is the whole lot worth?

	how many	each worth	together
High quality	60	\$10,000	\$600,000
Low quality	40	\$4,000	\$160,000
All	100		\$760,000

Sixty cars at ten thousand, forty at four thousand. Nothing here is more than multiplication and addition.

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Slide 13

Working through it

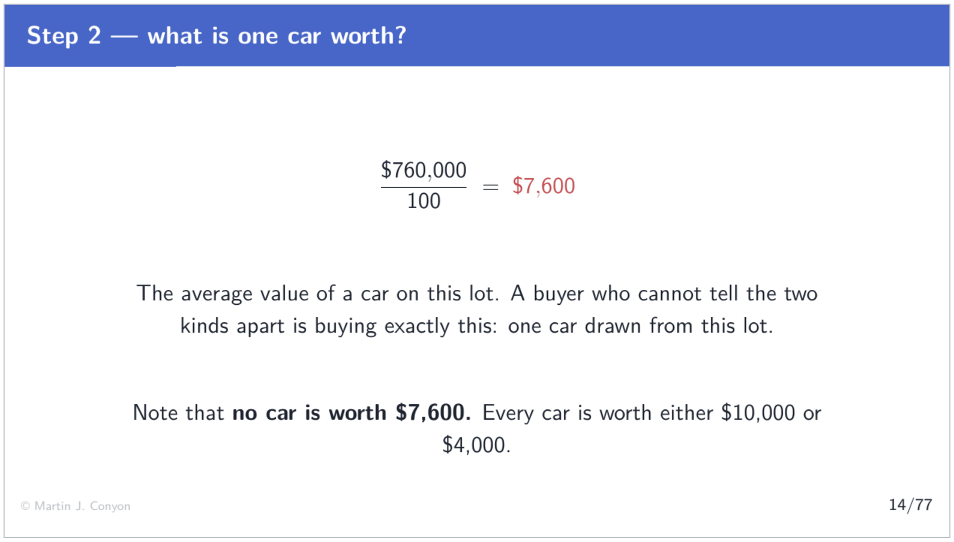
The arithmetic.

$$60 \times 10{,}000 = 600{,}000 \qquad 40 \times 4{,}000 = 160{,}000$$

$$600{,}000 + 160{,}000 = 760{,}000$$

What the number is. The total value to buyers of every car on the lot. It is not a price and nobody pays it. It is an intermediate quantity on the way to the answer.

Step 2 — what is one car worth?



Slide 14

Working through it

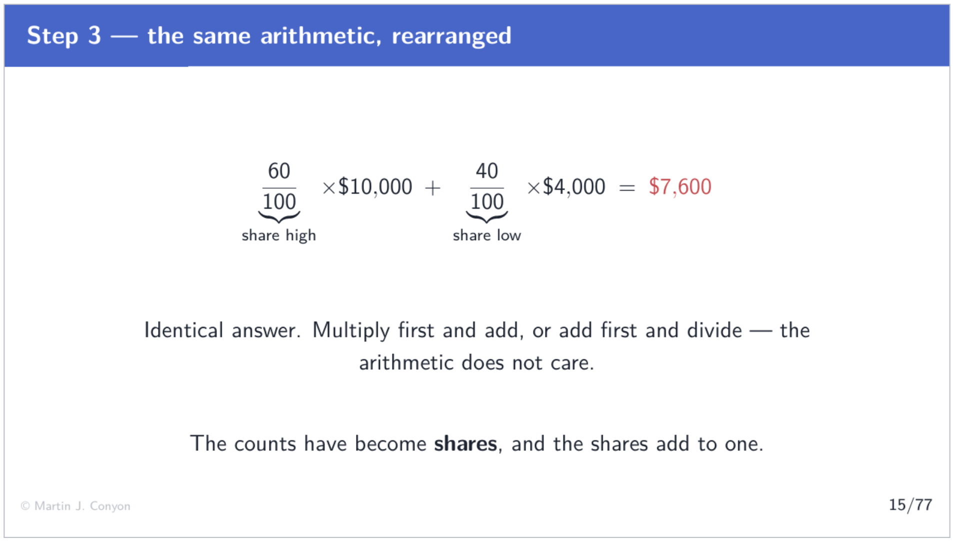
The division. $760{,}000 \div 100 = 7{,}600$.

What the number means. The average value across the lot. Equivalently: what a buyer would pay for a car chosen without knowing which kind it is, if they were paying exactly what it was worth to them on average.

The observation at the bottom is worth more than it looks. No individual car is worth $7,600. The average is a property of the collection. That is exactly why the market is about to run into trouble: a single price has to serve for two kinds of car, and it will be too low for one kind and too high for the other.

A check on the arithmetic. $7,600 lies between $4,000 and $10,000, and closer to $10,000, because most of the lot is high quality. Any answer outside that range would be wrong.

Step 3 — the same arithmetic, rearranged



Slide 15

Working through it

The rearrangement.

$$\frac{60}{100} \times 10{,}000 + \frac{40}{100} \times 4{,}000 = 0.6 \times 10{,}000 + 0.4 \times 4{,}000 = 6{,}000 + 1{,}600 = 7{,}600$$

Why the two routes agree. Dividing the total by 100 and multiplying each part by its share before adding are the same operation with the division moved earlier. Nothing has been assumed.

Why the shares add to one. Every car is high quality or low quality, each car was counted once, and there are a hundred of them. So $60 + 40 = 100$, and dividing through gives $0.6 + 0.4 = 1$. That is an observation about the counting, not an axiom.

Why this form is the useful one. It works when the counts are unknown but the shares are not. If sixty per cent of a national market is high quality, the same calculation runs without knowing how many cars there are.

Step 4 — a name and a symbol

Step 4 — a name and a symbol

Write λ for the share that is high quality. Then $1 - \lambda$ is the rest.

$$\text{average value} \;=\; \lambda\, v_B^H \;+\; (1-\lambda)\, v_B^L$$

This is called an **expected value**. It is the average of a list, written so that the shares appear instead of the counts.

That is the only sense in which the word "expected" is used in this lecture.

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Working through it

The notation. λ is the share of the lot that is high quality: 0.6 in the example, or sixty out of a hundred. $1 - \lambda$ is the remaining share.

Reading the formula. Each value is multiplied by the share of the lot it applies to, and the products are added:

$$\lambda\, v_B^H + (1-\lambda)\, v_B^L$$

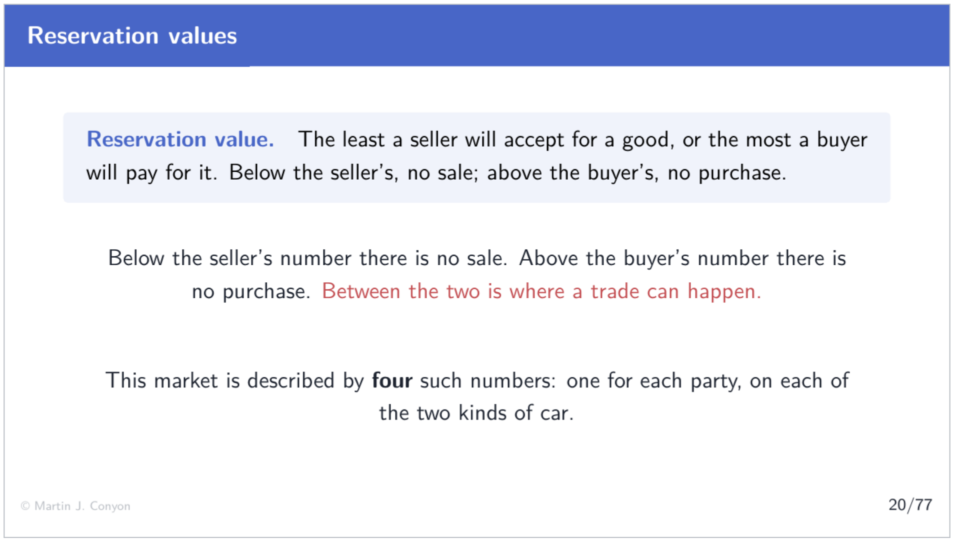
With $\lambda = 0.6$, $v_B^H = 10$ and $v_B^L = 4$ (in thousands), this is $0.6(10) + 0.4(4) = 7.6$.

On the word "expected". It is standard and slightly misleading. Nobody expects the value to be $7,600 — no car is worth that. The term is a technical name for a weighted average and nothing more.

The superscripts and subscripts. v_B^H reads: the value, to the Buyer, of a High-quality car. v_S^L: the value, to the Seller, of a Low-quality one. Four numbers in total, and Part 3 gives all four.

3. Part 3: Hidden Type: The Market for Used Cars

Reservation values



Slide 20

Working through it

What a reservation value is. The point at which a party is indifferent between trading and not. A seller's reservation value is what the good is worth to them if they keep it — its use to them, less any cost of continuing to hold it. A buyer's is what it is worth to them if they get it.

Why they differ, which is the ordinary reason any trade happens. The owner of a high-quality car has another car, or has moved somewhere with a train, or has simply had enough of it. It is worth $8,000 to them to keep. A buyer needs a car and it is worth $10,000 to them to have. Nothing mysterious.

The three regions. Take one car and one buyer. Below the seller's number, the seller refuses and there is no sale. Above the buyer's number, the buyer refuses and there is no purchase. Between the two, both are better off trading than not, and the price decides only how the gain is divided.

Why four numbers rather than two. There are two kinds of car and two parties, so each kind has a seller's number and a buyer's number. The next page gives all four.

The four numbers in this market

The four numbers in this market

	seller will not accept below	buyer will not pay above	gap
High quality	\$8,000	\$10,000	\$2,000
Low quality	\$2,000	\$4,000	\$2,000

A price inside the gap makes both parties better off. Every car here has such a gap.

Read the last column. The gap is **the same size on both rows**. No car in this market is a bad trade.

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Working through it

Reading the table.

- High quality: any price between \$8,000 and \$10,000 leaves both better off. At \$9,000 each gains \$1,000.
- Low quality: any price between \$2,000 and \$4,000 does the same. At \$3,000 each gains \$1,000.

The design of the example. The gap is deliberately the same on both types, \$2,000 each. That removes any suggestion that low-quality cars are less worth trading. Both types are equally worth trading, and the market is nonetheless about to destroy the trades in one of them.

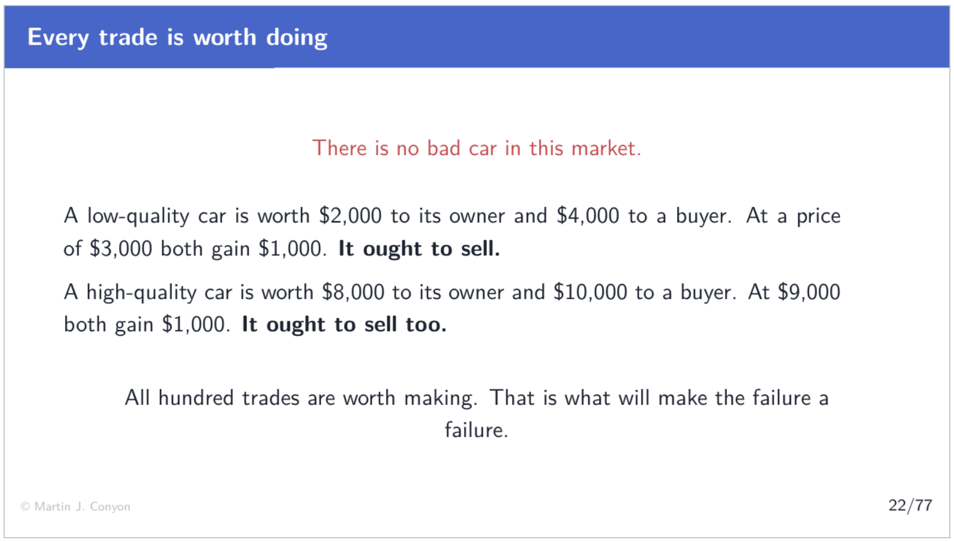
What the equal gaps rule out. If the low-quality gap were smaller, one could say the market destroys the trades that mattered least, and the failure would look less serious than it is. Here the two gaps are identical, so the trades that disappear are worth exactly as much as the ones that survive.

Notation for the four numbers.

$$v_S^H = 8, \quad v_B^H = 10, \quad v_S^L = 2, \quad v_B^L = 4 \qquad \text{(thousands)}$$

The subscript names the party — Seller or Buyer — and the superscript names the kind of car, High or Low.

Every trade is worth doing



Slide 22

Working through it

The low-quality trade, worked at $3,000.

- The seller gives up something worth $2,000 and receives $3,000, so gains $1,000.
- The buyer pays $3,000 for something worth $4,000, so gains $1,000.

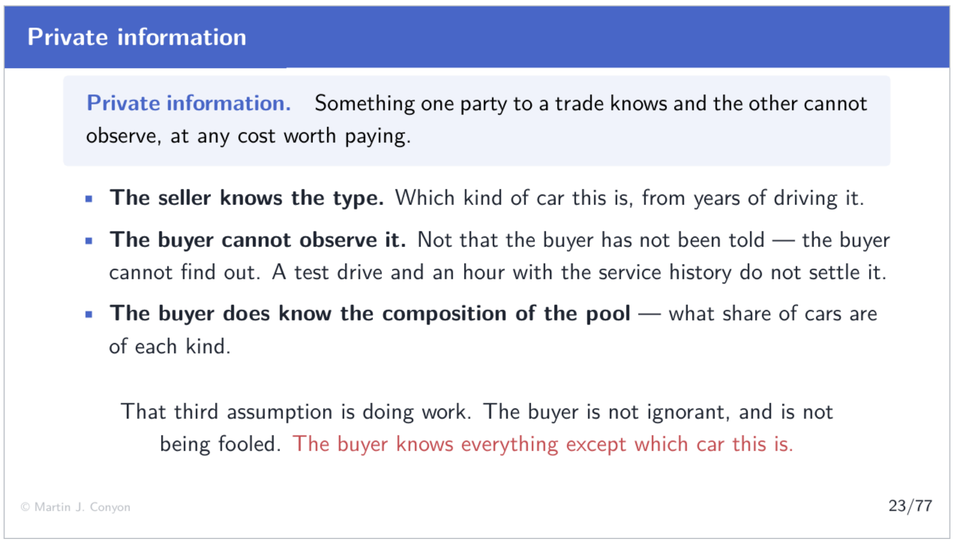
Total gain $2,000, split by where the price falls between the two reservation values.

Why this frame is here. The natural reading of the failure to come is "the bad cars drove out the good, and good riddance to the bad". That reading is wrong twice over. The market does not sort good from bad; and the trades it destroys are the *high-quality* ones, which were exactly as worth making as the others.

The benchmark, stated precisely. With quality observable, all hundred cars change hands and \$200,000 of surplus is created — \$2,000 on each of a hundred cars. Every number in Part 4 is measured against that figure.

What is not being assumed. No dishonesty anywhere. The seller is never required to lie and never does. The seller simply knows something the buyer does not.

Private information



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Working through it

The word that matters is unobservable. Private information is not information that has been withheld. It is information the other party cannot acquire at any cost worth paying. That is why the test drive and the service history are mentioned at all: they are the obvious ways of trying, and the model takes the case in which they fail.

Three assumptions, and the third is the one to notice.

The seller observes quality. Uncontroversial: they have owned the car.

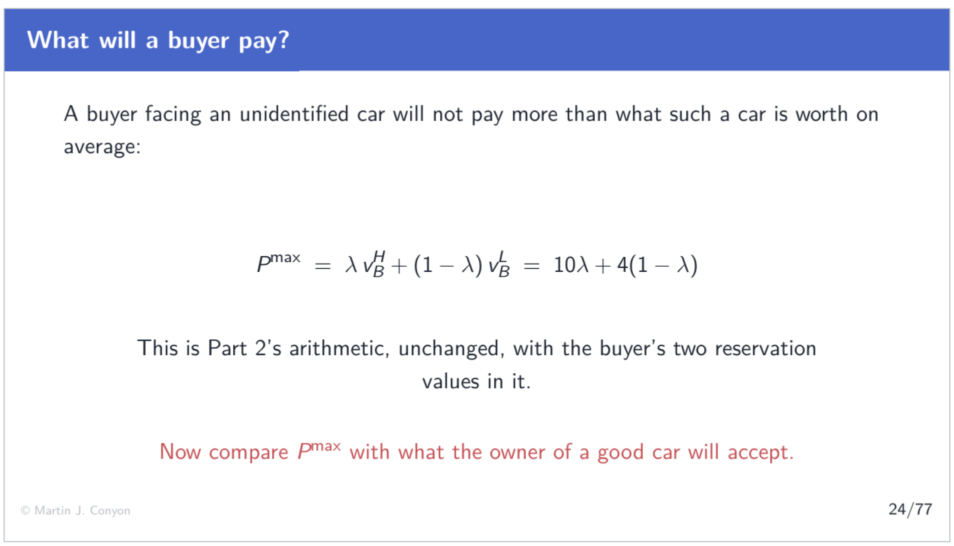
The buyer does not. Also reasonable for a car of any age. Some information is obtainable — an inspection, a history — and this model takes the case where it is not, so that the mechanism can be seen cleanly. Part 5 relaxes it.

The buyer knows the composition of the market. This is the assumption that makes the argument work, and it makes the buyer *sophisticated* rather than naive. The buyer cannot tell which car is in front of them, and can correctly work out what an unidentified car from this market is worth.

Why that matters for what follows. The failure is not caused by buyers being fooled. It happens with buyers who reason perfectly. That is what makes it a market failure rather than a story about gullibility: no amount of buyer sophistication fixes it, because sophistication is already assumed.

And it makes the failure worse, not better. A naive buyer might overpay and the good cars would keep selling. It is precisely because buyers correctly discount for what they cannot see that the good cars withdraw.

What will a buyer pay?



Slide 24

Working through it

Where the expression comes from. Part 2, with $v_B^H = 10$ and $v_B^L = 4$. Nothing new.

Why a buyer will not pay more. Paying above $P^{\max}$ means paying more than the car is worth on average, which loses money on average. A buyer who does that repeatedly goes broke.

Why a buyer might pay less, and why it does not matter. Bargaining could put the price anywhere between the seller's reservation value and $P^{\max}$. The argument to come only needs the *ceiling*: if even the highest price a buyer would pay is not enough to bring a good car to market, then no price is.

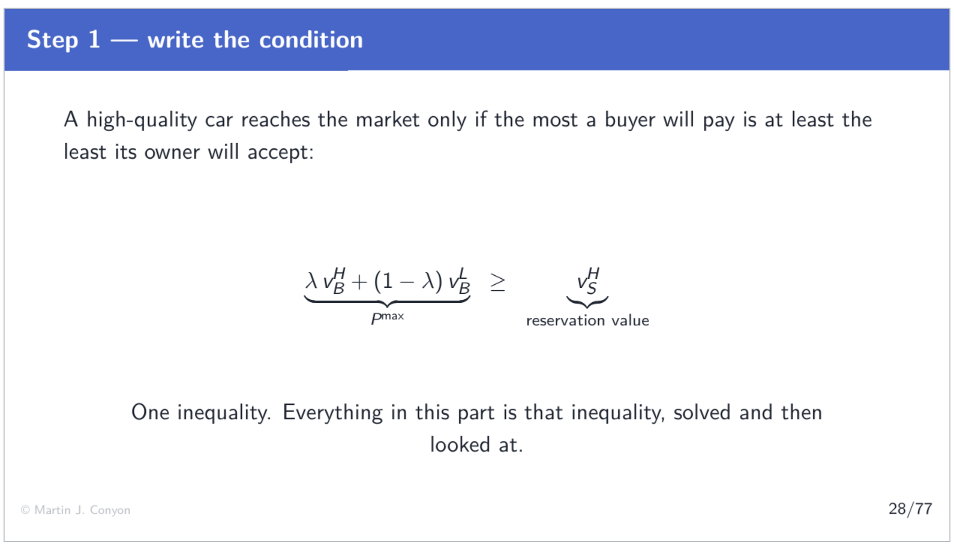
The comparison that decides everything. A high-quality owner will not accept below $v_S^H = 8$. A buyer will not pay above $P^{\max}$. So a high-quality car trades only if

$$P^{\max} \geq v_S^H$$

and Part 4 works out exactly when that holds.

4. Part 4: The Threshold

Step 1 — write the condition



Slide 28

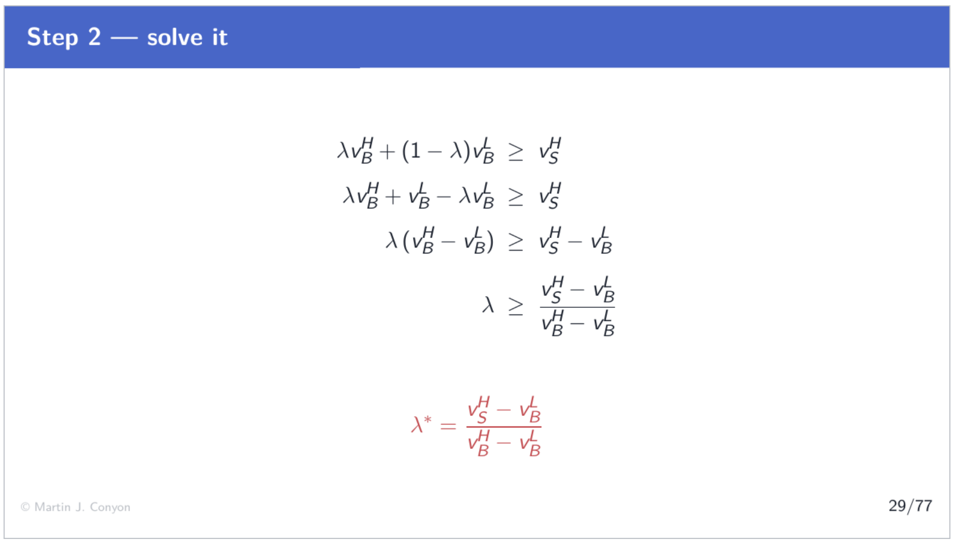
Working through it

Reading the two sides. The left is the ceiling on what a buyer will pay for a car they cannot identify. The right is the floor below which a high-quality owner will not sell. If the ceiling is below the floor there is no price both sides accept, and the trade does not happen.

Note which reservation values appear. The buyer's two values are on the left, because both kinds of car are in the pool the buyer is averaging over. The seller's *high* value is on the right, because that is the seller whose participation is in question. v_S^L does not appear at all, and Step 3 returns to why.

What is being asked. Not whether the market exists — low-quality cars will trade regardless. Whether the *good* cars stay in it.

Step 2 — solve it



Slide 29

Working through it

The algebra, line by line.

Multiply out: $\lambda v_B^H + v_B^L - \lambda v_B^L \geq v_S^H$.

Take v_B^L to the right: $\lambda v_B^H - \lambda v_B^L \geq v_S^H - v_B^L$.

Factor the left: $\lambda(v_B^H - v_B^L) \geq v_S^H - v_B^L$.

Divide by $v_B^H - v_B^L$, which is positive because a buyer values a good car above a bad one, so the inequality does not flip.

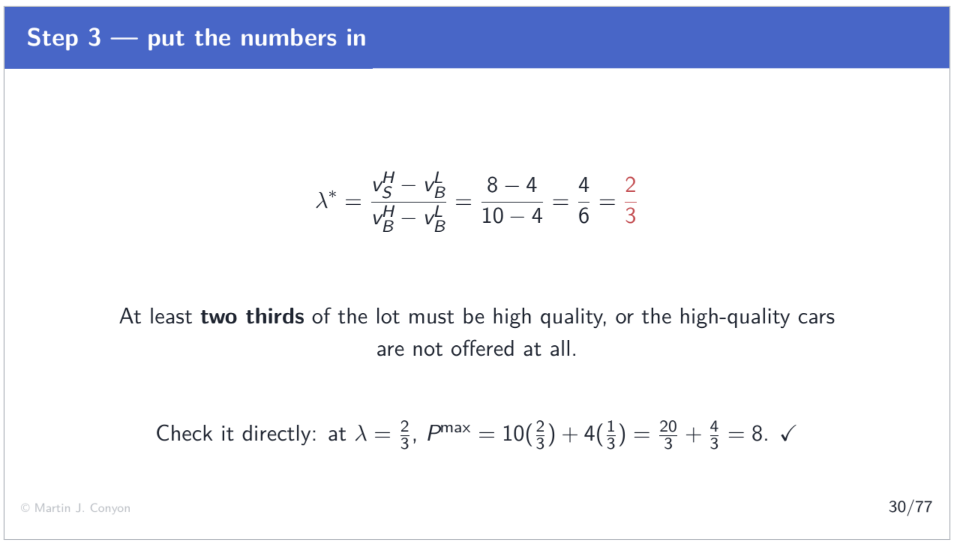
Reading the result before substituting anything.

$$\lambda^* = \frac{v_S^H - v_B^L}{v_B^H - v_B^L}$$

The numerator is the gap between what a good car's owner demands and what a buyer would pay for a bad one. The denominator is the spread in the buyer's own valuations. The threshold is the ratio of the two.

Why a general expression is worth the extra two lines. A number answers one market. An expression answers every market of this shape, and it shows which features matter. v_S^L is absent: what a bad car's owner thinks it is worth has no bearing on whether the good cars stay. That is not obvious in advance and falls straight out of the algebra.

Step 3 — put the numbers in



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Working through it

The substitution.

$$\lambda^* = \frac{8-4}{10-4} = \frac{4}{6} = \frac{2}{3} \approx 0.667$$

The direct check, which is worth doing every time. At $\lambda = \frac{2}{3}$:

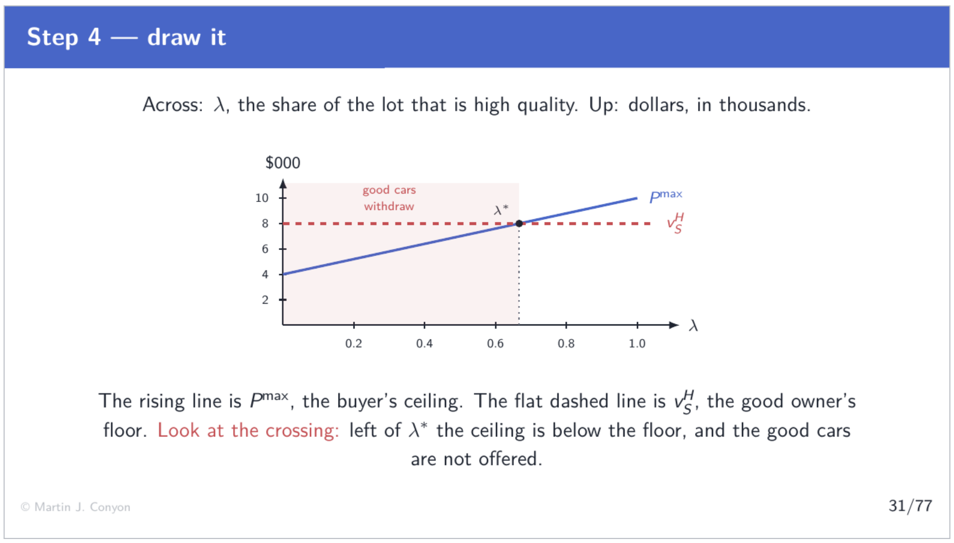
$$P^{\max} = 10\left(\tfrac{2}{3}\right) + 4\left(\tfrac{1}{3}\right) = \tfrac{20}{3} + \tfrac{4}{3} = \tfrac{24}{3} = 8$$

Exactly v_S^H. At the threshold the buyer's ceiling and the seller's floor coincide, which is what a boundary is.

What the number means in the market. Two out of every three cars must be high quality. If sixty-six per cent are, the good cars withdraw. If sixty-seven per cent are, they stay. The market's behaviour is discontinuous at that point.

And note what it is not. It is not a statement about how good the good cars are, nor about the size of the gains from trade. Both types have the same \$2,000 surplus. The threshold depends on the *levels* of the reservation values, not on the gaps.

Step 4 — draw it



Slide 31

Working through it

Reading the figure. λ on the horizontal axis, dollars in thousands on the vertical.

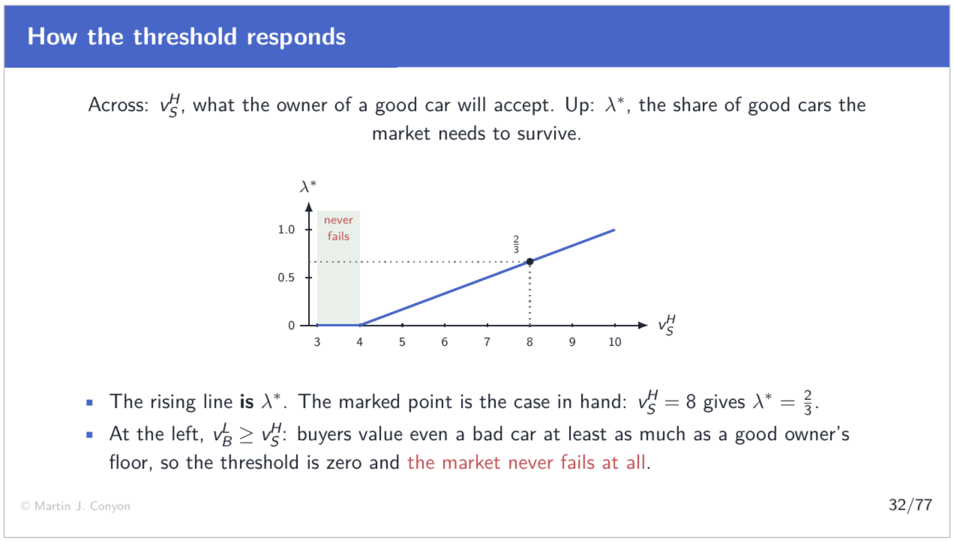
The rising line is $P^{\max} = 4 + 6\lambda$. At $\lambda = 0$ the lot is all low quality and a buyer will pay \$4,000. At $\lambda = 1$ it is all high quality and a buyer will pay \$10,000. It is a straight line because $P^{\max}$ is linear in λ, and its slope of 6 says that each extra share point of high quality adds \$60 to what a buyer will pay.

The horizontal line at \$8,000 is v_S^H, which does not depend on λ at all: what a good car is worth to its owner has nothing to do with what else is on the market.

The crossing is λ^*. Left of it the rising line is below the flat one and no price works; right of it the good cars trade.

Why the crossing is the whole answer. The threshold is where a rising line meets a flat one. Anything that raises the flat line, or lowers or flattens the rising one, moves the crossing right and makes the failure more likely. The next frame does exactly that.

How the threshold responds



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Working through it

What is plotted. v_S^H on the horizontal axis, λ^* on the vertical. The curve is the threshold expression itself, read as a function of one of its own inputs:

$$\lambda^*(v_S^H) = \frac{v_S^H - v_B^L}{v_B^H - v_B^L} = \frac{v_S^H - 4}{6}$$

a straight line of slope $\frac{1}{6}$, because the denominator does not move.

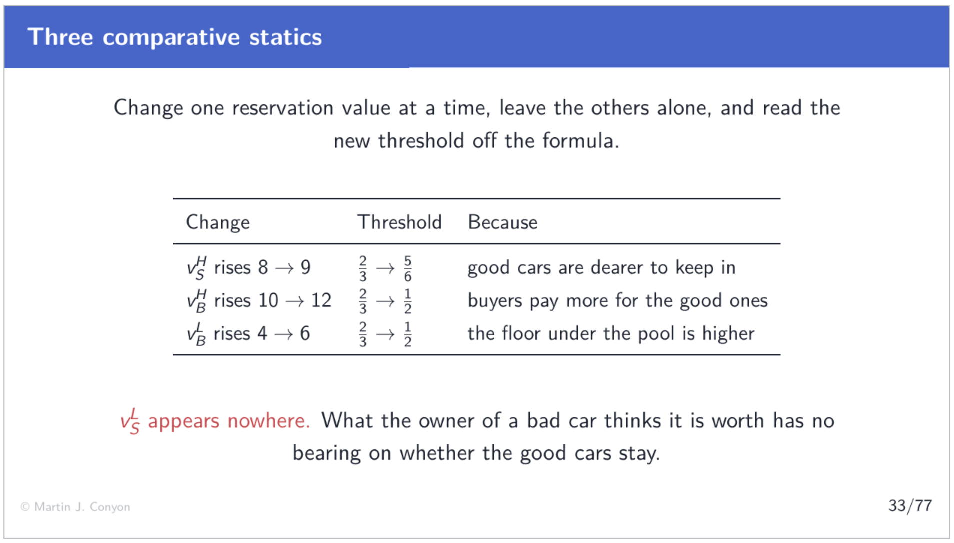
The marked point. At $v_S^H = 8$ the curve is at $(8-4)/6 = \frac{2}{3}$, which is the lecture's case.

Reading the slope. Each extra thousand dollars in what a good owner demands raises the required share of good cars by a sixth. The market becomes harder to sustain, and the reason is entirely one-sided: the owner's floor has risen and the buyer's ceiling has not.

The flat section at the left, and it is the important one. If $v_B^L \geq v_S^H$ then the numerator is zero or negative, so $\lambda^* \leq 0$ and the condition holds for every λ. Adverse selection cannot occur at all.

In words: if a buyer would pay more for a *bad* car than a good car's owner demands, then even the worst possible pool clears the good owner's floor, and nothing withdraws. **The failure requires the ranges to overlap** — the good seller's reservation value must sit above the buyer's valuation of a bad one. That is the condition under which this problem exists at all, and it is why adverse selection bites in some markets and not others.

Three comparative statics



Slide 33

Working through it

Three comparative statics, each computed from the same expression.

v_S^H **rises to 9.** $\lambda^* = (9-4)/(10-4) = \frac{5}{6}$. Now five sixths of the lot must be high quality. Better cars, in the sense of being worth more to their owners, are *harder* to keep in the market — the owner's floor has gone up and the buyer's ceiling has not.

v_B^H **rises to 12.** $\lambda^* = (8-4)/(12-4) = \frac{1}{2}$. The threshold falls. Buyers value good cars more, so the pooled price rises for any given composition.

v_B^L **rises to 6.** $\lambda^* = (8-6)/(10-6) = \frac{1}{2}$. Also falls, and this one is less obvious: raising the value of the *bad* cars raises the floor under the pooled price, which helps the good cars stay.

Why one at a time. Each row changes a single number and holds the other two fixed. That is what makes the *Because* column readable: with two numbers moving at once, the direction of the answer would depend on which moved further, and nothing would be learned.

What is absent. v_S^L appears nowhere in the expression, so it appears nowhere in the table. What the owner of a bad car thinks it is worth does not affect whether the good cars stay. That is not obvious in advance, and it falls straight out of the algebra rather than being assumed.

What it costs

What it costs

Take a lot that fails the condition: **60 high, 40 low**, so $\lambda = 0.6$ and $P^{\max} = \$7{,}600 < \$8{,}000$.

	quality observable	quality hidden
High-quality cars traded	60	0
Low-quality cars traded	40	40
Surplus created	\$200,000	\$80,000

$60 \times \$2{,}000 =$ \$120,000 of surplus destroyed

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Working through it

The arithmetic.

$$P^{\max} = 0.6(10) + 0.4(4) = 6 + 1.6 = 7.6 < 8$$

so the high-quality owners do not offer. Only the forty low-quality cars trade, at \$2,000 of surplus each, giving \$80,000 against a possible \$200,000.

$$60 \times \$2{,}000 = \$120{,}000$$

Sixty per cent of the achievable gains, lost. And a clear majority of the cars are good — sixty out of a hundred. A buyer still will not pay \$8,000, because of the forty that might turn up instead.

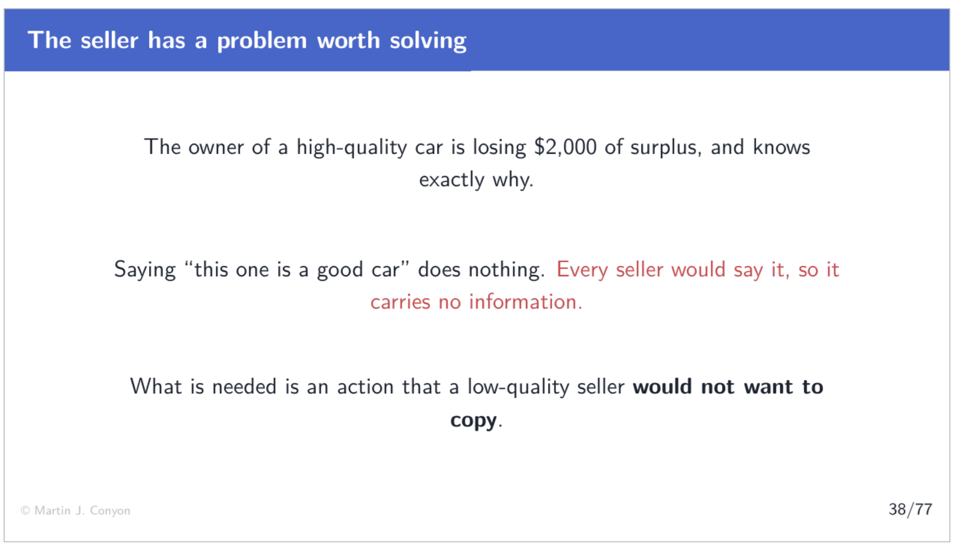
Who bears it. The sixty high-quality owners, who keep cars they would rather have sold, and the buyers who would have bought them. Both sides of sixty trades are worse off, and no third party gains. This is not a transfer; the surplus is not moved, it is destroyed.

What did not cause it. No monopoly: any number of buyers and sellers. No externality: no third party is affected. No dishonesty: no seller lies. No irrationality: buyers reason correctly from what they know. Remove every standard source of market failure and this one remains.

Why the number matters. It can be compared against the cost of a remedy. An inspection regime costing $500 a car is worth having if it recovers even a fraction of $120,000. Part 5 works out a remedy that costs the seller nothing at all when the car is genuinely good.

5. Part 5: Warranties: Proving What You Know

The seller has a problem worth solving



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Working through it

Why the problem has a solution at all. The informed party is the one losing. A high-quality owner is forgoing $2,000 of surplus and knows precisely why, which gives them both the motive and the knowledge to do something.

Why words do not work. A claim costs nothing to make. If saying "this car is high quality" raised the price, every seller would say it, buyers would learn that the statement carries no information, and the price would return to where it was. Economists call this **cheap talk**: a message with no cost attached conveys nothing, however sincere.

What is needed instead. An action that is *costly*, and — this is the whole of it — **differentially** costly. Not merely expensive, but more expensive for a seller with a bad car than for one with a good car. Then the fact that a seller has taken it is itself evidence.

The general name. This is a **signal**. The next three frames construct one and derive exactly when it works.

A warranty

A warranty

The seller offers: **if the car needs a major repair within a year, I pay you W.**

	repairs per hundred cars	expected cost of the warranty
High quality	10	0.10 W
Low quality	60	0.60 W

The same promise costs a low-quality seller **six times** what it costs a high-quality one.

That difference is what makes it a signal rather than talk.

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Working through it

The promise. A commitment to pay W if the car needs a major repair within a year. Note that it is a commitment made *before* the buyer can verify anything, and enforceable afterwards.

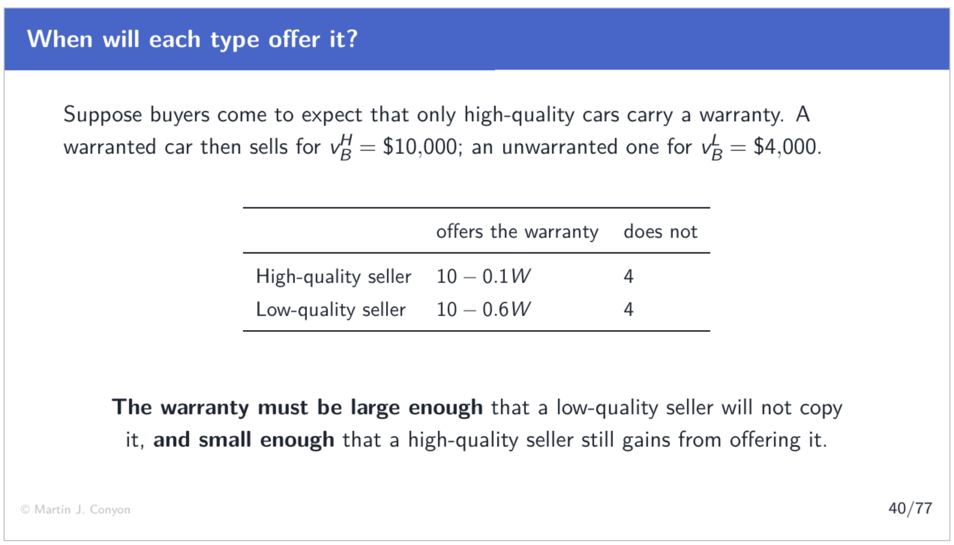
The repair rates, as counts. Of a hundred high-quality cars, 10 need a major repair in the first year. Of a hundred low-quality cars, 60 do. These are facts about the cars, and the seller knows which kind is being sold.

The expected cost, using Part 2's arithmetic. A high-quality seller expects to pay out on 10 cars in every 100, so the warranty costs $\frac{10}{100}W = 0.1W$. A low-quality seller expects to pay out on 60 in 100, so it costs $0.6W$.

The ratio is what matters. $0.6W/0.1W = 6$. The identical promise is six times as expensive for the seller of a bad car. Nothing about the warranty itself distinguishes the two — the words are the same. What differs is what the promise will cost to keep, and that is determined by the very thing the buyer cannot observe.

Why this is the natural remedy. The signal is tied to the hidden characteristic mechanically. A seller cannot make a bad car cheap to warrant, and that is precisely why offering the warranty is informative.

When will each type offer it?



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Working through it

The setup. We are looking for a **separating** outcome: high-quality sellers offer the warranty, low-quality sellers do not, and buyers correctly infer quality from the offer.

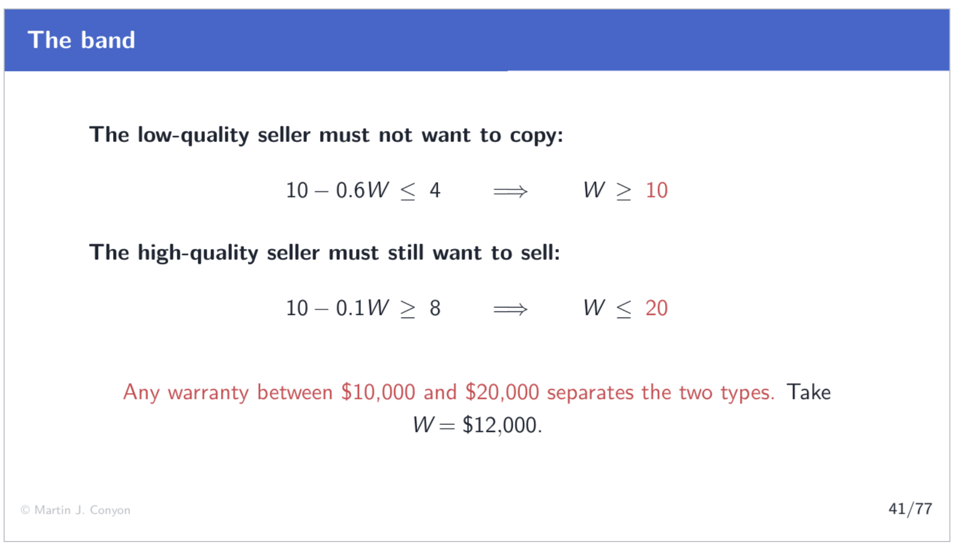
Why the prices are what they are. If buyers believe the warranty separates the types, a warranted car is known to be high quality and fetches up to $v_B^H = 10$. An unwarranted one is known to be low quality and fetches up to $v_B^L = 4$.

The table, row by row.

- A high-quality seller offering the warranty receives 10 and expects to pay $0.1W$, netting $10 - 0.1W$.
- A low-quality seller copying receives 10 and expects to pay $0.6W$, netting $10 - 0.6W$.
- Either, selling without a warranty, receives 4.

Note the circularity, and why it is not a problem. The prices depend on buyers' beliefs, and buyers' beliefs are correct only if the sellers behave as expected. That is what an equilibrium is: beliefs and behaviour that are consistent with each other. The next frame checks that a W exists for which they are.

The band



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Working through it

The first condition, in full. A low-quality seller copying the warranty receives 10 and expects to pay $0.6W$. Not copying gives 4. So copying is unattractive when

$$10 - 0.6W \leq 4 \implies 6 \leq 0.6W \implies W \geq 10$$

The second condition. A high-quality seller offering the warranty receives 10 and expects to pay $0.1W$, and will only do so if that beats the reservation value of 8:

$$10 - 0.1W \geq 8 \implies 0.1W \leq 2 \implies W \leq 20$$

Both together.

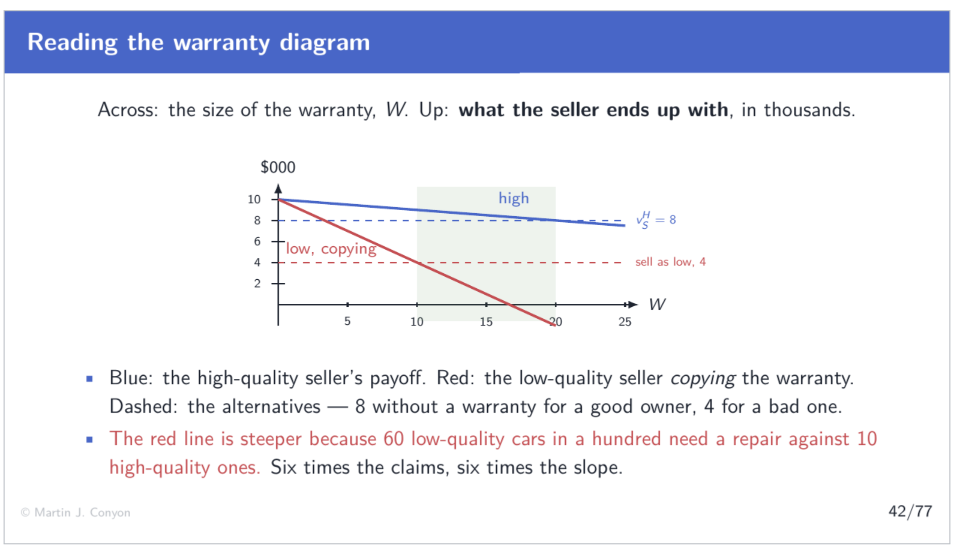
$$10 \leq W \leq 20$$

The band is non-empty, so a separating warranty exists. Its width, \$10,000, is the room available.

Reading the two edges. The lower edge is set by the *bad* type's temptation to copy, and by the difference in repair rates. The upper edge is set by the *good* type's own reservation value and by how much buyers will pay.

The condition for a band to exist at all. There is one if the lower bound is below the upper bound. Had the two repair rates been closer together — say 40 and 50 per hundred rather than 10 and 60 — the warranty needed to deter the bad type would be so large that the good type would refuse it too, and no signal would work.

Reading the warranty diagram



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Working through it

What the vertical axis measures. Not the price, and not the warranty. It is what the seller *ends up with*: the price received, less the expected cost of honoring the promise.

The two payoff lines. Both start at 10 when $W = 0$, because with no warranty both would receive the warranted price. They then fall as W rises:

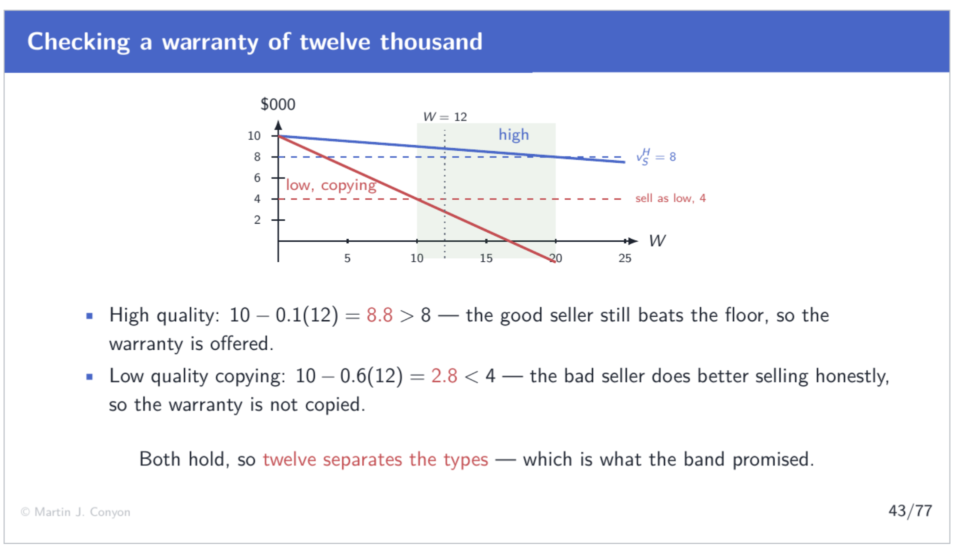
$$\text{high: } 10 - 0.1W \qquad \text{low, copying: } 10 - 0.6W$$

and **the low type's line falls six times as steeply**, because the expected payout is six times larger. That single fact is what the whole figure is about.

The two floors. The high type's alternative is keeping the car, worth $v_S^H = 8$. The low type's alternative is selling without a warranty at $v_B^L = 4$. They are different alternatives, and that is why the two dashed lines are at different heights.

Reading the shaded band. It runs from where the red line crosses 4 — at $W = 10$ — to where the blue line crosses 8 — at $W = 20$. Inside it the blue line is above its floor and the red line is below its own, which is exactly the pair of conditions written down on the previous page.

Checking a warranty of twelve thousand



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Working through it

The check at $W = 12$.

high: $10 - 0.1(12) = 10 - 1.2 = 8.8 > 8$ low copying: $10 - 0.6(12) = 10 - 7.2 = 2.8 < 4$

The dotted vertical line sits at 12. Where it meets the blue line is 8.8, above the blue dashed floor. Where it meets the red line is 2.8, below the red dashed floor. Both conditions are visible as well as arithmetical.

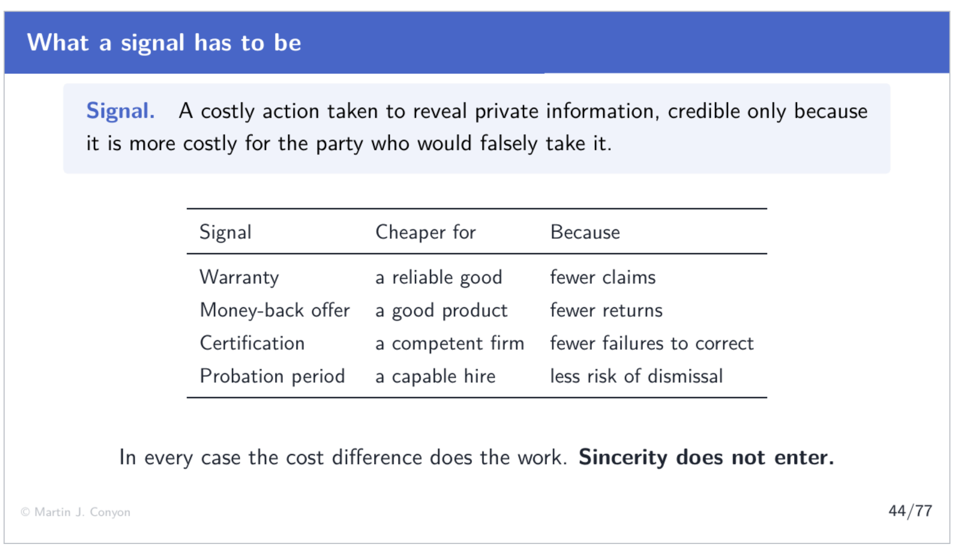
What each check decides. The first decides whether the good seller *offers* the warranty. The second decides whether the bad seller *copies* it. A separating outcome needs both answers, and 12 gives them.

What the high-quality seller has achieved. Without a warranty, at $\lambda = 0.6$, the car did not sell at all. With one, the seller nets \$8,800 — \$800 above the reservation value. The \$1,200 expected cost of the warranty buys back \$2,000 of surplus that was otherwise lost.

And what the buyer has achieved. A car known to be high quality, plus insurance against the ten-in-a-hundred chance of a repair.

Why any W in the band would do. Nothing singles out 12. At $W = 15$ the high type nets $10 - 1.5 = 8.5$, still above 8, and the low type copying nets $10 - 9 = 1$, still below 4. The band is the answer; 12 is one point in it.

What a signal has to be



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Working through it

The condition, stated once. A costly action separates types when the cost is *lower* for the type that genuinely has the characteristic. Then taking the action is evidence of the characteristic, and buyers can rely on it without trusting anyone.

Why cost alone is not enough. An action that is expensive for everyone equally separates nothing: either both types take it or neither does. It is the *difference* in cost that carries the information.

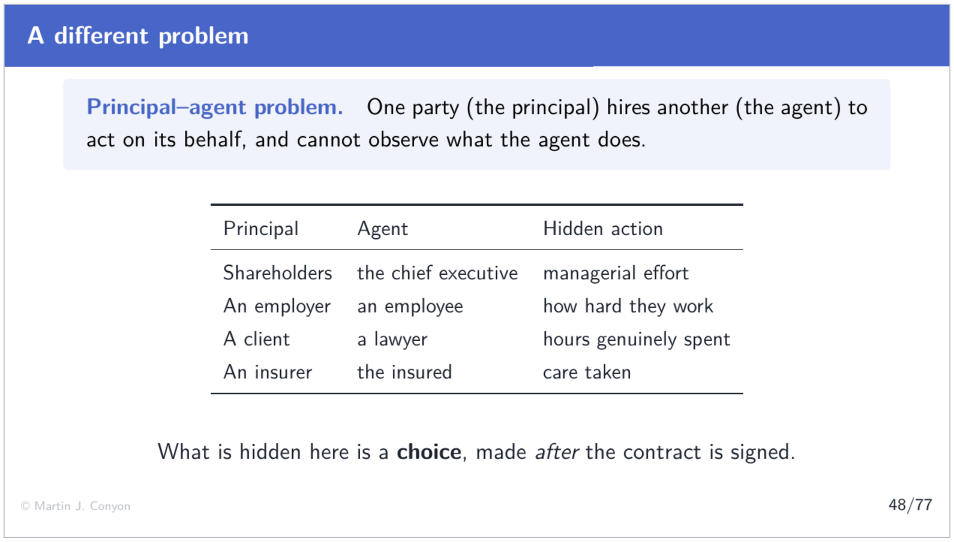
Reading the table. In each row, the signal is mechanically tied to the hidden characteristic. A firm cannot make its unreliable product cheap to guarantee. The tie is what makes the signal credible.

Note what is absent from the argument. No appeal to honesty, reputation, morality, or long-run relationships. The low-quality seller in this model would copy the warranty at once if it paid — and does not, because it does not pay. That is a stronger result than one relying on good behaviour, because it holds whatever the seller's character.

The general name. This is **signalling**: the informed party takes a costly action to reveal what it knows. The mirror image is **screening**, where the *uninformed* party designs choices that induce revelation — an insurer offering a menu of deductibles, where the low-risk applicant takes the high deductible and the high-risk one does not.

6. Part 6: Hidden Action: The Principal and the Agent

A different problem



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Working through it

The structure. Two parties. The principal wants something done and delegates it. The agent does it, and chooses how much effort to supply. The principal cannot observe that choice.

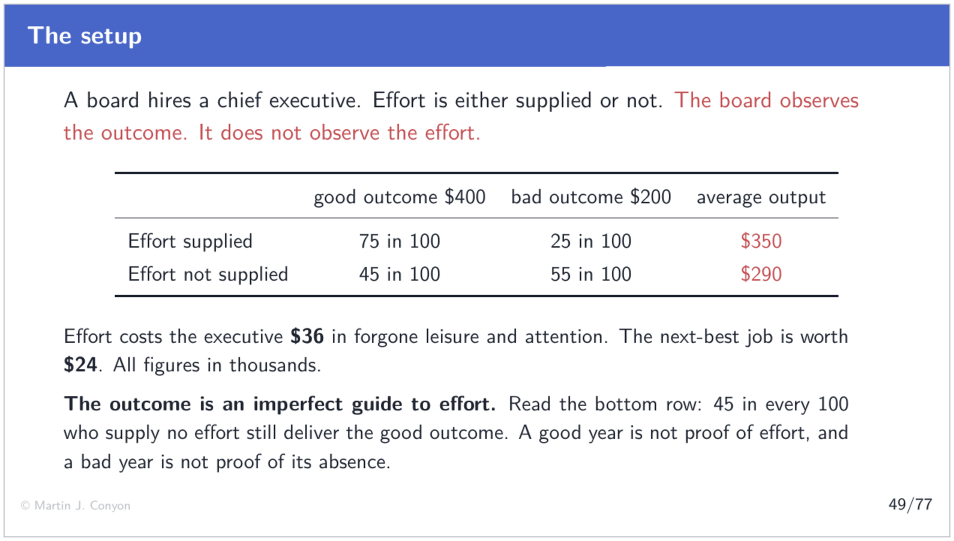
The contrast with Part 3, which is the whole distinction. There the hidden thing was fixed before the parties met, and no contract could alter it. Here the hidden thing does not exist until the agent supplies it, and a contract signed now can influence it.

Why that changes what can be done. Against hidden type, the informed party signals. Against hidden action, the principal writes a contract that makes the agent *want* to take the right action. Different problem, different remedy.

Why it is not simply solved by watching. Effort is not the same as attendance. A manager can be present, busy, and supplying very little of what the firm actually needs. What the principal can verify — hours, presence, activity — is not what the principal wants, and what the principal wants is not verifiable.

The term "moral hazard" and why it is unhelpful. It comes from nineteenth-century fire insurance and suggests bad character. Nothing in the analysis requires it. An agent responding to the incentives in a contract is behaving exactly as economics assumes everyone behaves.

The setup



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Working through it

Everything as counts. Seventy five in a hundred, not a probability of nought point seven five. Same fact, no apparatus required.

The average outputs, from Part 2.

$$\text{with effort: } 0.75(400) + 0.25(200) = 300 + 50 = 350$$

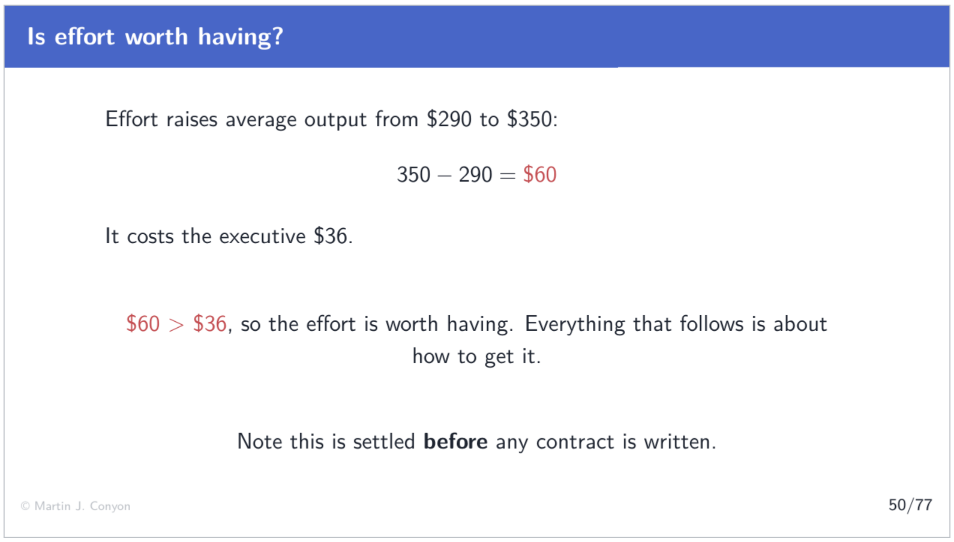
$$\text{without: } 0.45(400) + 0.55(200) = 180 + 110 = 290$$

Note what effort does and does not do. It does not guarantee the good outcome — twenty five in a hundred hard-working executives still deliver 200. It shifts the odds. That is realistic and it is also what makes the problem hard: a bad outcome is not proof that effort was withheld.

The two numbers describing the executive. Effort costs \$36, which is what the executive gives up in leisure and attention. The outside option is \$24, which is what the next-best job is worth — below that, the executive does not sign.

What the board can and cannot see. It sees the outcome, 400 or 200. It does not see the effort. So a contract cannot say "supply effort and be paid"; it can only say "if the outcome is 400 you are paid this, and if 200 you are paid that".

Is effort worth having?



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Working through it

The comparison. The gain from effort is the rise in average output, $350 - 290 = 60$. The cost is what the executive gives up, 36. Since $60 > 36$, a total surplus of 24 is available from the effort being supplied.

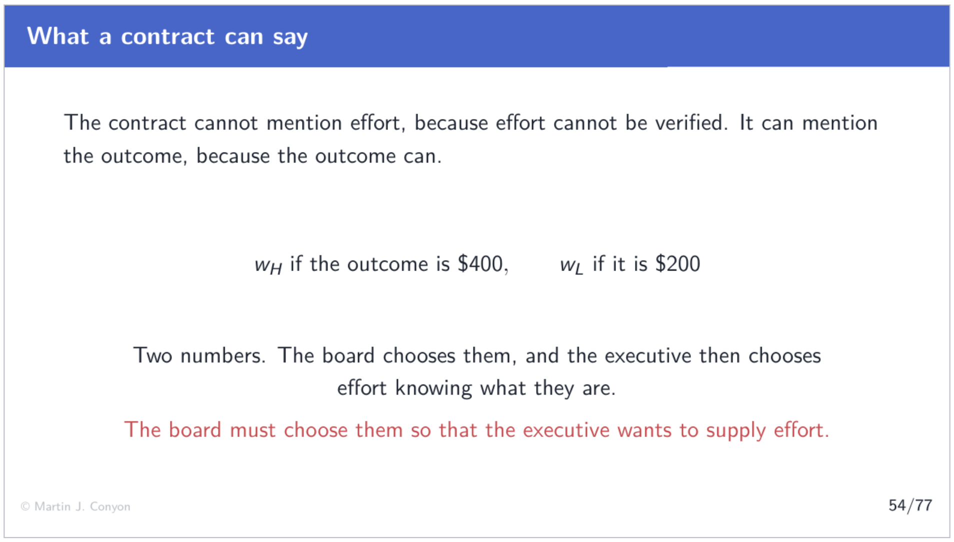
Why this comes first. If effort cost more than it produced, the efficient outcome would be no effort, and the contracting problem would be trivial: pay a flat wage and let the executive take it easy. The interesting case is the one where effort is worth having *and* cannot be observed.

The benchmark this establishes. If effort were observable, the board would write "supply effort" into the contract, pay the executive $24+36 = 60$ — the outside option plus compensation for the effort — and keep $350-60 = 290$. That is the first best, and Part 8 measures against it.

Why the board cannot simply do that. Writing "supply effort" into a contract requires a court to be able to determine whether it happened. Nobody can. The clause is unenforceable, so it is worthless.

7. Part 7: The Two Conditions a Contract Must Satisfy

What a contract can say



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Working through it

The constraint that shapes everything. A contract can only condition on things a court could verify. Effort is not one of them. Output is.

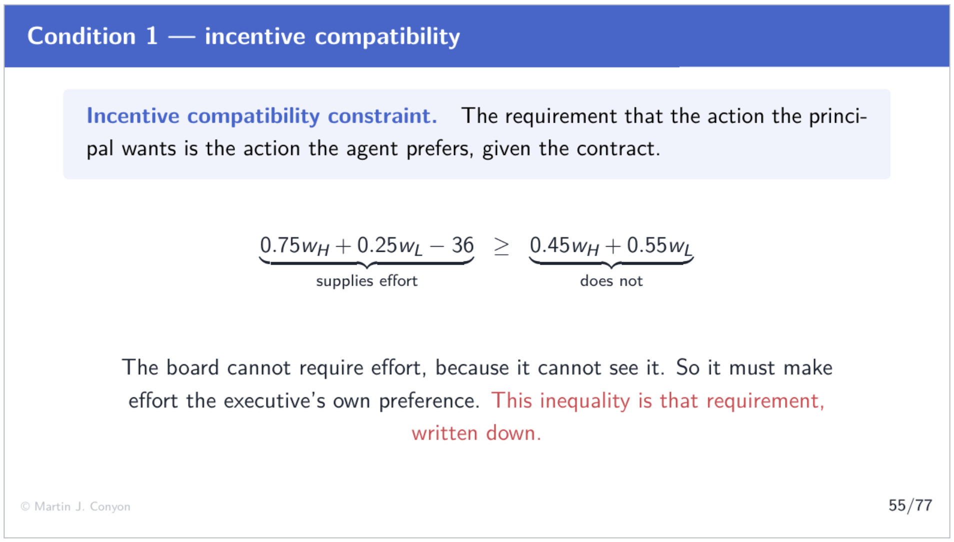
So the contract is a pair of numbers. w_H paid when output is 400, w_L paid when it is 200. Nothing else.

The order of moves. The board sets w_H and w_L. The executive sees them and chooses whether to supply effort. Then the outcome occurs and the wage is paid. The board is choosing first, knowing how the executive will respond.

Why paying for output works at all, given that output is not effort. Because effort shifts the odds: 75 in a hundred rather than 45. So an executive paid more when output is high has a reason to supply effort, even though the link is imperfect.

And why it is imperfect in a way that matters. An executive who supplies effort may still deliver 200 and be paid w_L. The contract punishes bad luck along with low effort, because it cannot tell them apart. Part 9 returns to what that costs.

Condition 1 — incentive compatibility



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Working through it

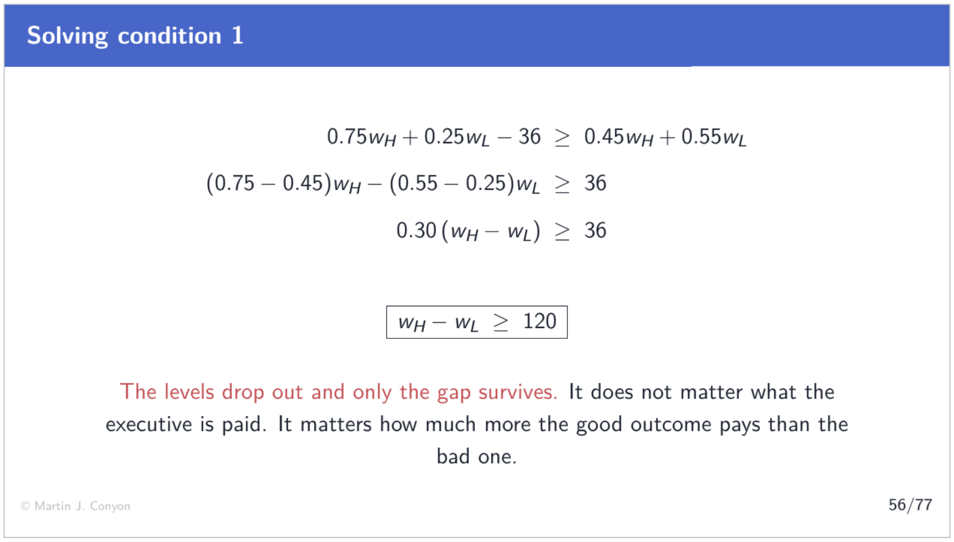
The name. The **incentive compatibility constraint**: a contract is incentive compatible when the action the principal wants is the action the agent would choose anyway, given the terms. It is the formal version of a plain idea — you cannot order what you cannot verify, so you must arrange for it to be wanted.

The comparison. With effort, the executive receives w_H in 75 cases out of a hundred and w_L in 25, and bears the cost 36. Without effort, they receive w_H in 45 cases and w_L in 55, and bear no cost. Both sides are weighted averages, built exactly as in Part 2.

Why the left side carries the -36 *and the right side does not.* The cost of effort is incurred only when effort is supplied. It is a cost of the action, not of the job.

What the inequality does not say. It does not say the executive is honest, or diligent, or well disposed to the board. It says only that, given these two numbers, supplying effort pays at least as well as not. The next page solves it.

Solving condition 1



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Working through it

The algebra, line by line.

Start from the comparison and take every wage term to the left:

$$0.75w_H - 0.45w_H + 0.25w_L - 0.55w_L \geq 36$$

which is

$$(0.75 - 0.45)w_H - (0.55 - 0.25)w_L \geq 36$$

The step that makes the result. Both differences are the same number, 0.30 — necessarily so, because the two columns of counts each add to a hundred, so whatever effort adds to the good outcome it takes from the bad one. That is why the expression factors:

$$0.30w_H - 0.30w_L = 0.30\,(w_H - w_L) \geq 36 \implies w_H - w_L \geq 120$$

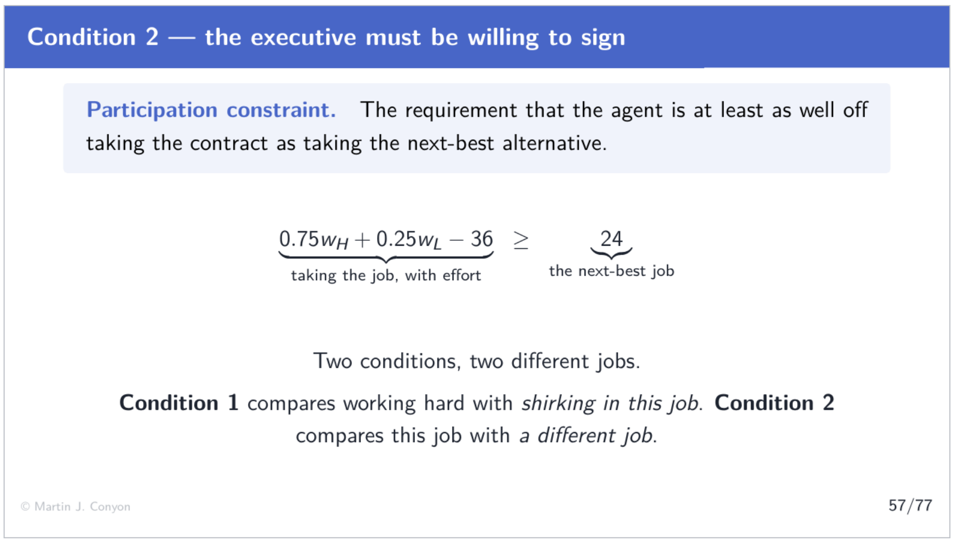
Reading the result. The *levels* of the two wages have dropped out entirely. Only the gap matters. Adding \$50 to both wages leaves the incentive to supply effort exactly unchanged, because it is paid in both states.

The general form, which is worth having.

$$w_H - w_L \geq \frac{\text{cost of effort}}{\text{how much effort shifts the odds}} = \frac{36}{0.30} = 120$$

Effort that costs more requires a larger gap. Effort that shifts the odds less — a smaller denominator — also requires a larger gap, because the outcome is weaker evidence of what was supplied.

Condition 2 — the executive must be willing to sign



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Working through it

The inequality. The executive's position on taking the job and supplying effort is $0.75w_H + 0.25w_L - 36$. The alternative is the outside option, 24. The job must be at least as good.

Why two conditions rather than one. They rule out different failures.

Condition 1 fails $\Rightarrow$ the executive signs and does not supply effort. The board has an employee who is not doing the thing they were hired for.

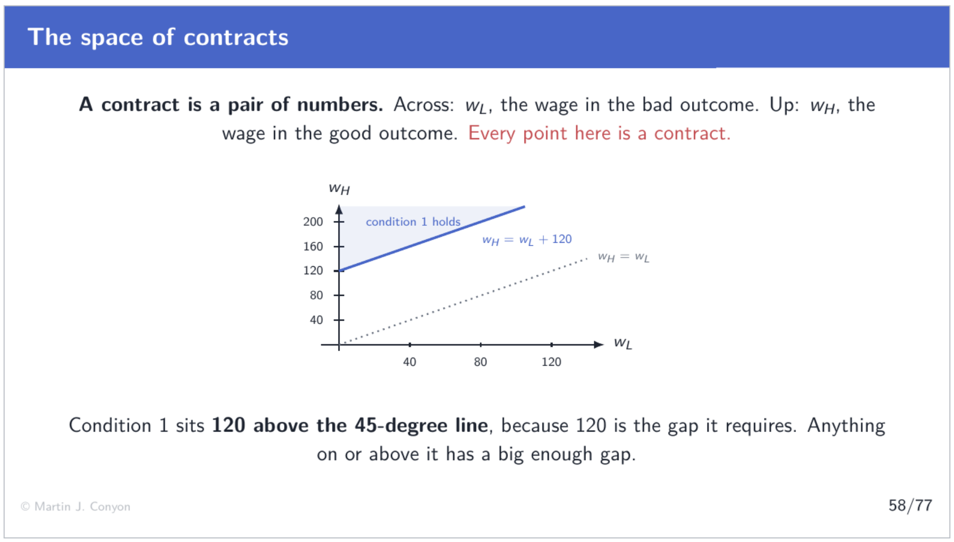
Condition 2 fails $\Rightarrow$ the executive does not sign. The board has no employee.

A contract must satisfy both, and they pull in the same direction — both are easier to satisfy with higher pay — but they constrain different features of the pay. Condition 1 constrains the *gap*; condition 2 constrains the *level*.

The standard name. The **participation** condition, or the individual rationality condition.

Which one binds. That depends on the numbers, and it is the question Part 8 turns on. When the participation condition binds, the executive earns exactly the outside option. When the incentive condition binds, the executive earns more than it — and the difference is what the board pays for not being able to observe effort.

The space of contracts



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Working through it

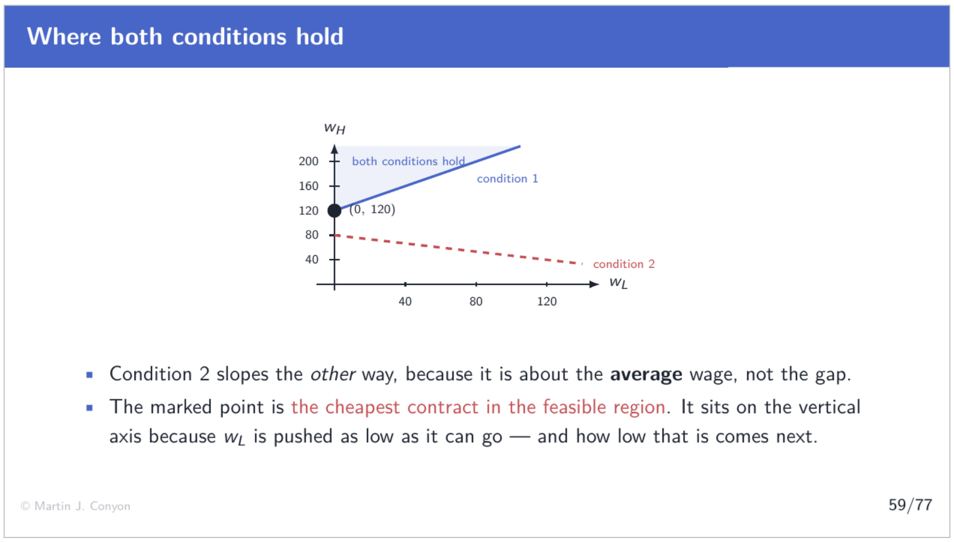
Why the picture is worth drawing. A contract here is two numbers, and any pair of numbers is a point in a plane. So the set of all possible contracts is the whole quadrant, and a condition on a contract is a region within it. Nothing is being approximated: the picture is the algebra.

Reading the axes. w_L horizontal, w_H vertical.

The 45-degree line. Along it $w_H = w_L$: the two outcomes pay the same, so the gap is zero. It is drawn only as a reference, because the position of condition 1 is read against it.

Condition 1 as a region. $w_H - w_L \geq 120$ means $w_H \geq w_L + 120$. That is everything on or above a line of slope 1 with vertical intercept 120 — the 45-degree line lifted by exactly the gap the condition demands. The vertical distance between the two lines is 120 everywhere, which is why the picture makes the condition legible: the gap is a height, and it is the same height at every w_L.

Where both conditions hold



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Working through it

Condition 2 as a region. $0.75w_H + 0.25w_L \geq 60$, so $w_H \geq (60 - 0.25w_L)/0.75$. A downward-sloping line, because a higher wage in the bad state allows a lower one in the good state while keeping the executive willing to sign. It constrains the *level* of pay; condition 1 constrains the *gap*. That is why the two lines slope in opposite directions.

The feasible set. Everything above both lines. Any contract in it induces effort and is accepted. In this market condition 1 is the higher of the two throughout the quadrant, so the feasible set is simply the region above condition 1 — which is itself the result Part 8 turns on.

Why the board does not stop here. Every contract in the region works, and they do not all cost the same. The board's expected wage bill is $0.75w_H + 0.25w_L$, so it wants the cheapest point in the region.

Where that is. Along the condition 1 boundary, $w_H = w_L + 120$, the bill is

$$0.75(w_L + 120) + 0.25 w_L = w_L + 90$$

which falls as w_L falls. So the board drives w_L down as far as it is allowed to, and the marked point $(0, 120)$ is where it stops. What stops it is the subject of Part 8.

8. Part 8: Limited Liability, the Cost, and Executive Pay

First, a case where nothing is lost

First, a case where nothing is lost

Suppose the executive can be made the **residual claimant**: buy the firm for a fixed fee F and keep whatever it earns.

Outcome \$400	the executive keeps $400 - F$
Outcome \$200	the executive keeps $200 - F$
The gap	$400 - 200 = 200$, far above the 120 required

Condition 2 then fixes F: the board's cost is $24 + 36 = \$60$, exactly the outside option plus the cost of effort.

Hidden action costs nothing here.

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Working through it

The arrangement. The executive pays a fixed fee F and keeps the firm's output. Wages become $w_H = 400 - F$ and $w_L = 200 - F$.

Why the incentive condition is satisfied easily. The gap is $(400 - F) - (200 - F) = 200$, which does not depend on F at all. It is well above the 120 required, because the executive now bears the full consequence of the outcome.

The fee, from condition 2. The executive's expected position is $350 - F - 36$, and this must reach 24, giving $F = 290$. The board receives $F = 290$ and the executive nets $350 - 290 - 36 = 24$, exactly the outside option. The board's total cost is $350 - 290 = 60 =$

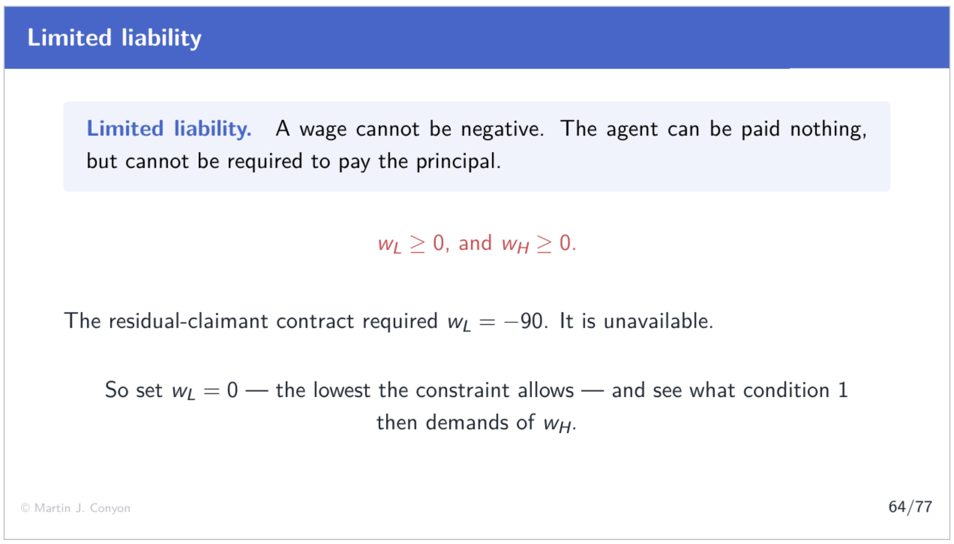
$24 + 36$.

This is the first best. The board pays only what is unavoidable: the executive's outside option plus compensation for the effort. Nothing is lost to the information problem.

And this is a real arrangement. The owner-operator, the franchisee, the taxi driver who owns the medallion, the law-firm partner. Each is a case of the agent being made the residual claimant, and each is a response to exactly this problem.

What it requires, and this is where it breaks. With $F = 290$, the wage in the bad state is $200 - 290 = -90$. The executive must be able to lose \$90. Owning a business means bearing the loss, and most managers cannot and will not.

Limited liability



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Working through it

The constraint. $w_L \geq 0$ and $w_H \geq 0$. Employees are not required to pay their employers.

Why it is not merely a legal technicality. Even where a negative payment could be written, an executive without \$90 to hand cannot make it, and one who has it would demand compensation for the risk of losing it. The constraint binds for almost every employee of almost every firm.

What it rules out. Precisely the contract that solved the problem. The residual-claimant arrangement worked by making the executive bear the downside, and limited liability is the statement that the downside cannot be borne.

The board's remaining choice. Set w_L as low as allowed — zero — and use w_H to satisfy the incentive condition. Raising w_L would only require raising w_H by the same amount to preserve the gap, and would cost more. So $w_L = 0$ is where the cheapest contract sits.

Where this is heading. With w_L pinned at zero, the gap is w_H itself, and condition 1 sets a floor under it. The next frame checks whether that floor is above or below what condition 2 requires.

Both conditions with the bad-state wage at zero

Both conditions with the bad-state wage at zero

Set $w_L = 0$. Limited liability allows nothing lower, and every dollar of w_L would have to be matched in w_H to hold the gap.

Condition 1. The gap is $w_H - 0 = w_H$, so

$$w_H \geq 120$$

Condition 2.

$$0.75w_H - 36 \geq 24 \implies 0.75w_H \geq 60 \implies w_H \geq 80$$

The 0.75 is there because 75 of every 100 executives who supply effort deliver the good outcome — so that is the share of the time w_H is actually paid.

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Working through it

Why $w_L = 0$ and not something higher. Limited liability allows nothing lower, and nothing higher is worth doing. Raising w_L by a dollar forces w_H up by a dollar as well, or the gap of 120 is lost and the incentive condition fails. The board would then be paying more in both states for exactly the same behavior.

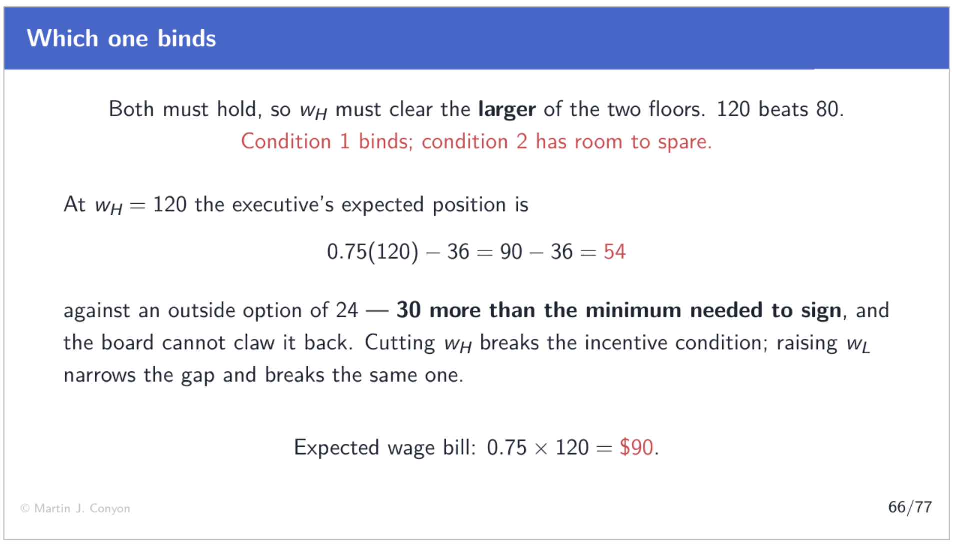
Condition 1 with $w_L = 0$. The gap is $w_H - 0 = w_H$, so the condition reads $w_H \geq 120$ directly.

Condition 2 with $w_L = 0$*.* The executive is paid w_H in 75 cases out of a hundred and nothing in the other 25, so the expected wage is $0.75w_H$, and the position after the cost of effort is $0.75w_H - 36$:

$$0.75w_H - 36 \geq 24 \implies 0.75w_H \geq 60 \implies w_H \geq 80$$

Where the 0.75 *comes from.* It is the count from the setup: 75 of every 100 executives who supply effort deliver the good outcome. That is the share of the time w_H is actually paid, so it is the weight w_H carries in the executive's expected pay.

Which one binds



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Working through it

What binds *means.* Both conditions must hold at once, so w_H must be at least the larger of the two floors they impose. The larger floor is the one doing the work — it is said to **bind** — and the other is *slack*: satisfied, but not determining anything. Here 120 exceeds 80, so condition 1 binds and condition 2 is slack.

Why that is the interesting case. If condition 2 bound, the executive would be paid exactly enough to sign and no more, and the information problem would cost the board nothing. It is condition 1 binding that forces the board past what it would otherwise have to pay.

The consequence, which is the point of the part. At $w_H = 120$ the executive's expected position is

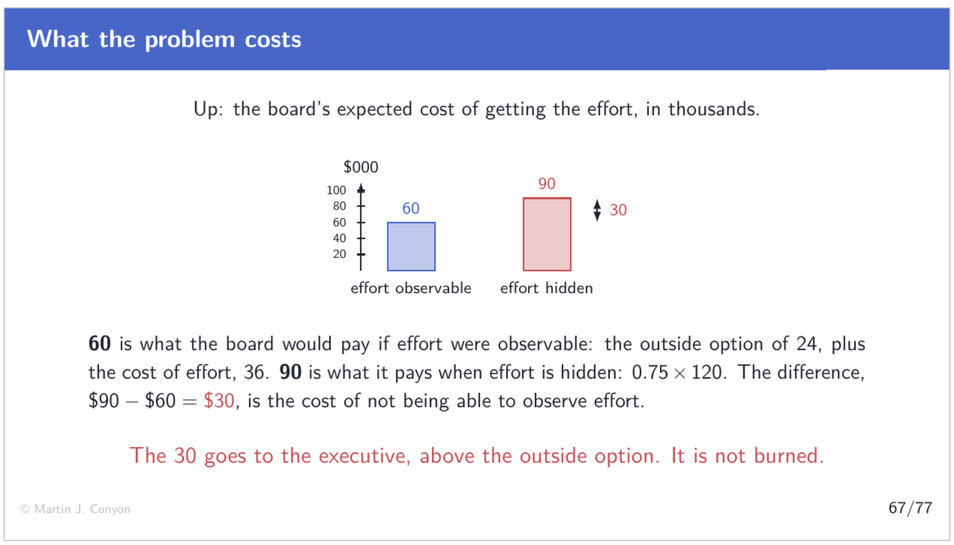
$$0.75(120) - 36 = 90 - 36 = 54$$

against an outside option of 24. The executive is **30 better off than the minimum needed to sign**.

Why the board cannot claw that back. Reducing w_H below 120 breaks the incentive condition and the executive stops supplying effort. Paying part of it in the bad state — raising w_L — narrows the gap and breaks the same condition. The surplus the executive collects is not a bargaining outcome. It is forced by the two constraints together.

The expected cost to the board. $0.75 \times 120 + 0.25 \times 0 = 90$.

What the problem costs



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Working through it

The two numbers.

$$\text{effort observable: } 24 + 36 = 60 \qquad \text{effort hidden: } 0.75 \times 120 = 90$$

$$\text{agency cost} = 90 - 60 = 30$$

Where the 30 goes. To the executive. Expected pay of 90, less the effort cost of 36, is 54 against an outside option of 24 — a surplus of exactly 30. The agency cost and the executive's surplus are the same 30, seen from the two sides.

So is anything actually lost? In this model, no: the 30 is transferred rather than destroyed. That is a consequence of assuming the executive is indifferent to risk. Once agents dislike risk — Part 9 — the transfer becomes a genuine loss, because the executive must additionally be compensated for bearing an outcome they do not control.

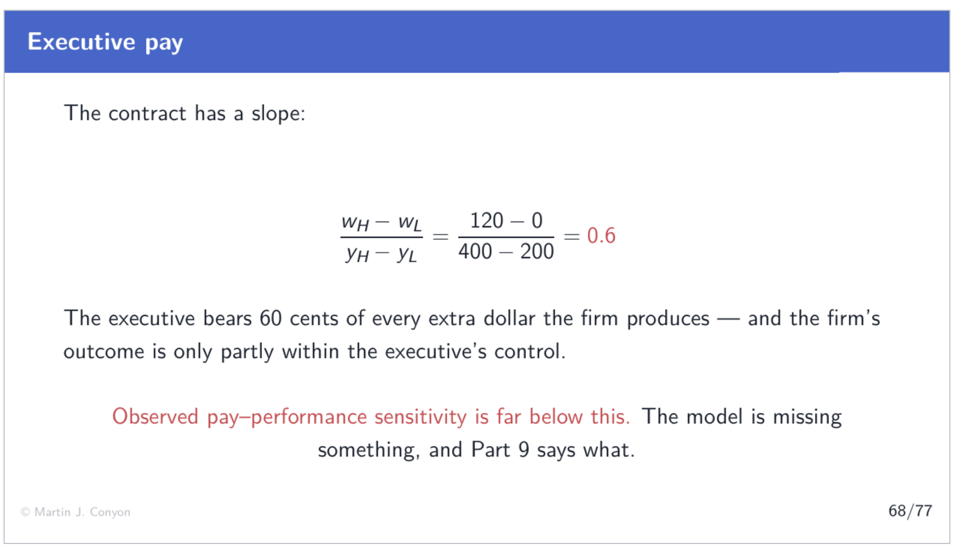
Why the board cannot recover it. Every route is blocked. Lower w_H and the incentive condition fails. Raise w_L to claw back some of it and the gap narrows, so w_H must rise by the same amount, leaving the board worse off. Charge the executive an up-front fee and limited liability blocks it.

The general expression, which is worth having. The gap required is

$$w_H - w_L = \frac{\text{cost of effort}}{\text{shift in the odds}}$$

so anything that raises the cost of effort, or weakens the link between effort and outcome, raises the required gap and with it the agency cost.

Executive pay



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Working through it

The slope. A pay contract can be described by how much pay changes per dollar of output:

$$\frac{w_H - w_L}{y_H - y_L} = \frac{120}{200} = 0.6$$

What determines it. Substituting the general form of the required gap:

$$\text{slope} = \frac{\text{cost of effort}}{(\text{shift in the odds}) \times (y_H - y_L)} = \frac{36}{0.30 \times 200} = 0.6$$

Effort that is costly to supply requires a steeper contract. Effort that shifts the odds less also requires a steeper one, because the outcome is weaker evidence about what was supplied.

The empirical mismatch. Estimated sensitivities for large public firms are small — typically a few dollars of executive wealth per thousand dollars of shareholder value. The model here says 60 cents in the dollar. The gap is large and it is informative.

What the model is missing. Risk. This executive is indifferent to it, so a steep contract costs the board nothing extra. A real executive dislikes bearing an outcome driven substantially by interest rates, commodity prices and the business cycle, and must be paid to bear it. That payment rises with the slope, so the optimal contract trades incentives against insurance and settles far flatter.

Why the question is never "should pay depend on performance". It has to, or condition 1 fails and no effort is supplied. The question is how steeply, and that is a trade-off rather than a principle.

9. Part 9: What the Model Leaves Out

Four assumptions doing real work

Four assumptions doing real work

- **Two types, two outcomes, two effort levels.** Real quality and real effort are continuous. The direction of every result survives; the clean thresholds do not.
- **Nobody minds risk.** The whole of Part 8's transfer becomes a genuine loss once the agent must be paid to bear an outcome they do not control.
- **One period, one transaction.** No reputation, no repeat business, no future in which a name is worth protecting.
- **A fixed set of instruments.** A price and a warranty in the first half; two wages in the second.

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Working through it

Two types. The smallest number that makes the problem interesting, and it gives a clean threshold in λ. With a continuum of qualities the withdrawal is not a single step: the best cars leave, which lowers the average, which causes the next-best to leave, and so on. Akerlof's original paper works that case, and the failure is worse rather than different.

Risk neutrality, which is the most consequential omission. Part 8's agency cost of 30 was a transfer from the board to the executive, not a destruction of value. That is only true because the executive is indifferent to risk. A risk-averse executive facing a contract with a slope of 0.6 must be compensated for bearing an outcome driven by the business cycle, and that compensation is a genuine loss. The optimal contract then trades incentives against insurance, and settles far flatter — which is why observed pay-performance sensitivity is a small fraction of what this model implies.

One period. A used-car dealer who will be in business next year has a name worth protecting, and reputation can substitute for a warranty. Nothing in this model has a future in it, so nothing here can capture that.

A fixed set of instruments. The remedies available were fixed in advance. Widen the set and other responses appear: inspections and third-party certification against hidden type; deferred pay, promotion and dismissal against hidden action.

Two remedies, one principle

Two remedies, one principle

	Signalling	**Screening**
Who acts	the informed party	the uninformed party
What they do	take a costly action	offer a menu of choices
Works when	the cost differs by type	the choice differs by type
Example	a warranty	insurance deductibles

Part 5 built a signal. Screening is its mirror image, and rests on the same condition: the two types must find the same option differently attractive.

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Working through it

Signalling. The informed party moves first, taking an action whose cost depends on the hidden characteristic. Part 5's warranty, education as a signal of ability, a firm submitting to costly certification.

Screening. The uninformed party moves first, offering a set of options designed so that different types choose differently. The classic case is insurance: offer a high-deductible policy at a low premium and a low-deductible policy at a high one. A low-risk applicant, expecting few claims, takes the high deductible. A high-risk applicant does not. The insurer learns the type from the choice without ever observing it.

The shared condition. Both work only when the two types rank the options differently. If a bad car's owner found the warranty exactly as cheap as a good car's owner, the warranty would separate nothing. If both risk types preferred the same policy, the menu would separate nothing.

Why the distinction is worth keeping. It tells you where to look for a remedy. If the informed party is losing, expect signalling. If the uninformed party is losing, expect screening. Used-car sellers offer warranties; insurers offer deductibles.

Where this leads

Where this leads

- **Insurance.** Both problems at once: the applicant knows the risk (hidden type) and chooses the care taken (hidden action).
- **Credit.** Lenders cannot tell good borrowers from bad, and raising the interest rate worsens the pool — so lending is rationed rather than priced.
- **Labor.** Education as a signal, probation as screening, performance pay against hidden action.
- **Corporate governance.** Boards, auditors and disclosure rules exist because owners cannot observe what managers do.

Each is the same two problems, in a different market.

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Working through it

Insurance. Applicants know their own risk, which the insurer does not: hidden type. Once insured, they choose how much care to take, which the insurer also cannot observe: hidden action. Deductibles address both — they screen at the point of sale and preserve some incentive for care afterwards.

Credit. A lender facing borrowers of unknown quality might expect to raise the interest rate to compensate. Raising it drives out the safest borrowers, whose projects cannot bear it, leaving a riskier pool — the used-car mechanism exactly. The result is that lenders *ration* credit rather than clearing the market with price, which is one of the few settled results in economics where a market does not clear.

Labor. Education as a signal of ability, on the argument that studying is less costly for the more able. Probation periods as screening. Performance pay, bonuses and equity against hidden action.

Corporate governance. Boards, external auditors, disclosure requirements and independent directors are all responses to the impossibility of shareholders observing managerial action directly. The design of executive pay — how steep, and against what measure — is Part 8's problem as an applied question.